

Lab 1

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11:59PM February 1

You should have RStudio installed to edit this file. You will write code in places marked “TO-DO” to complete the problems. Most of this will be a pure programming assignment but there are some questions that instead ask you to “write a few sentences”. This is a W class! The tools for the solutions to these problems can be found in the class practice lectures. I prefer you to use the methods I taught you. If you google and find esoteric code you don’t understand, this doesn’t do you too much good.

To “hand in” the homework, you should first download this file. The best way to do this is by cloning the class repository then copying this file from the folder of that clone into the folder that is your personal class repository. Then do the assignment by filling in the TO-DO’s. After you’re done, compile this file into a PDF (use the “knit to PDF” button on the submenu above). This PDF will include output of your code. Then push the PDF and this Rmd file by the deadline to your github repository in a directory called “labs”.

Basic R Skills

- Print out the numerical constant pi with ten digits after the decimal point using the internal constant pi.

```
options(digits=11)
pi
```

```
## [1] 3.1415926536
```

- Sum up the first 103 terms of the series $1 + 1/2 + 1/4 + 1/8 + \dots$

```
sum(2^(-(0:102)))
```

```
## [1] 2
```

- Find the product of the first 37 terms in the sequence $1/3, 1/6, 1/9 \dots$

```
prod(1/(3*(1:37)))
```

```
## [1] 1.613528728e-61
```

- Find the product of the first 387 terms of $1 * 1/2 * 1/4 * 1/8 * \dots$

```
prod(2^(-(0:386)))
```

```
## [1] 0
```

Is this answer *exactly* correct?

No, because none of the terms in the product is equal to 0. This is an example of numerical “underflow.”

We use sums of logarithms to help with products.

- Figure out a means to express the answer more exactly. Not compute exactly, but express more exactly.

```
sum(log10(2^(-(0:386))))
```

```
## [1] -22484.231406
```

- Create the sequence `x = [Inf, 20, 18, ..., -20]`.

```
c(Inf, seq(20, -20, by = -2))
```

```
## [1] Inf 20 18 16 14 12 10 8 6 4 2 0 -2 -4 -6 -8 -10 -12 -14
## [20] -16 -18 -20
```

Create the sequence `x = [log_3(Inf), log_3(100), log_3(98), ... log_3(-20)]`.

```
x = log(c(Inf, seq(100, -20, by = -2)), base=3)
```

```
## Warning: NaNs produced
```

Comment on the appropriateness of the non-numeric values `NAN` and `-Inf`.

- Create a vector of booleans where the entry is true if `x[i]` is positive and finite.

```
x > 0 & is.finite(x)
```

```
## [1] FALSE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
## [13] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
## [25] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
## [37] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
## [49] TRUE TRUE TRUE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
## [61] FALSE FALSE
```

- Locate the indices of the non-real numbers in this vector. Hint: use the `which` function. Don't hesitate to use the documentation via `?which`.

```
?which
```

```
which(is.infinite(x) | is.nan(x))
```

```
## [1] 1 52 53 54 55 56 57 58 59 60 61 62
```

- Locate the indices of the infinite quantities in this vector.

```
which(is.infinite(x))
```

```
## [1] 1 52
```

- Locate the indices of the min and max in this vector. Hint: use the `which.min` and `which.max` functions.

```
y = x
```

```
y[is.infinite(x)] = NA
```

```
which.min(y)
```

```
## [1] 51
```

```
which.max(y)
```

```
## [1] 2
```

- Count the number of unique values in `x`.

```
length(unique(x))
```

```
## [1] 53
```

- Cast `x` to a factor. Do the number of levels make sense?

```
as.factor(x)
```

```
## [1] Inf 4.19180654857877 4.1734172518943 4.15464876785729
## [5] 4.13548512895119 4.11590933734319 4.09590327428938 4.07544759935851
## [9] 4.05452163806914 4.03310325630434 4.01116871959141 3.98869253500376
## [13] 3.96564727304425 3.94200336638929 3.91772888178973 3.89278926071437
## [17] 3.86714702345081 3.84076143030548 3.8135880922156 3.78557852142874
## [21] 3.75667961082847 3.72683302786084 3.69597450568212 3.6640330098758
## [25] 3.63092975357146 3.59657702661571 3.56087679500731 3.52371901428583
## [29] 3.48497958377173 3.44451784578705 3.40217350273288 3.3577627814323
## [33] 3.31107361281783 3.26185950714291 3.20983167673402 3.15464876785729
## [37] 3.09590327428938 3.03310325630434 2.96564727304425 2.89278926071437
## [41] 2.8135880922156 2.72683302786084 2.63092975357146 2.52371901428583
## [45] 2.40217350273288 2.26185950714291 2.09590327428938 1.89278926071437
## [49] 1.63092975357146 1.26185950714291 0.630929753571457 -Inf
## [53] NaN NaN NaN NaN
## [57] NaN NaN NaN NaN
## [61] NaN NaN
## 53 Levels: -Inf 0.630929753571457 1.26185950714291 ... NaN
```

- Cast `x` to integers. What do we learn about R's infinity representation in the integer data type?

```
as.integer(x)
```

```
## Warning: NAs introduced by coercion to integer range
## [1] NA 4 4 4 4 4 4 4 4 4 4 3 3 3 3 3 3 3 3 3 3 3 3
## [26] 3 3 3 3 3 3 3 3 3 3 3 3 3 2 2 2 2 2 2 2 2 1 1 1
## [51] 0 NA NA NA NA NA NA NA NA NA NA NA NA
```

- Use `x` to create a new vector `y` containing only the real numbers in `x`.

```
y = x[is.finite(x)]
```

- Use the left rectangle method to numerically integrate x^2 from 0 to 1 with rectangle width size $1e-6$.

```
sum((seq(0,1-(1e-6), by = 1e-6))^2)*(1e-6)
```

```
## [1] 0.33333283333
```

- Calculate the average of 100 realizations of standard Bernoullis in one line using the `sample` function.

```
?sample
mean(sample(c(0,1),size=100, replace=TRUE))
```

```
## [1] 0.52
```

- Calculate the average of 500 realizations of Bernoullis with $p = 0.9$ in one line using the `sample` and `mean` functions.

```
mean(sample(c(0,1), size=500, replace=TRUE, prob=c(0.1,0.9)))
```

```
## [1] 0.884
```

- Calculate the average of 1000 realizations of Bernoullis with $p = 0.9$ in one line using `rbinom`.

```
mean(rbinom(n=1000, size=1, prob=0.9))
```

```
## [1] 0.905
```

- Let $n = 50$. Create a $n \times n$ matrix `R` of exactly 50% entries 0's, 25% 1's 25% 2's. These values should be in random locations.

```
vec = sample(c(rep(0,1250), rep(1,625), rep(2,625)))
vec
```

```
##      [1] 1 1 2 0 1 1 1 0 0 2 0 1 0 0 0 2 2 2 1 0 1 2 0 1 1 1 0 0 2 1 2 2 0 0 0 2 0
##     [38] 0 1 0 0 1 0 0 0 1 2 0 1 0 0 0 1 2 0 0 0 0 0 0 0 2 2 2 0 0 0 0 2 0 0 0 0 1
##     [75] 0 0 0 0 2 0 2 0 0 2 1 0 0 0 0 0 0 0 2 0 0 0 0 0 2 0 2 0 0 2 0 0 0 0 1 0
##    [112] 0 0 0 1 1 2 1 0 2 0 0 2 0 1 1 2 2 0 0 1 1 1 2 0 0 0 0 1 1 0 1 1 2 0 1 2 0
##    [149] 1 2 1 2 2 2 2 2 0 2 0 1 2 0 0 0 1 2 2 0 1 0 0 0 2 0 1 0 0 0 0 0 2 0 0 2 1
##    [186] 0 1 0 2 0 1 2 2 1 2 0 0 1 0 0 2 0 1 1 0 0 1 0 0 1 2 0 2 0 1 0 1 0 1 0 1 2
##    [223] 0 0 0 2 0 0 2 2 0 0 1 0 1 0 0 0 0 2 1 1 2 0 1 2 0 0 2 0 0 2 1 1 0 0 0 2 2
##    [260] 0 0 0 0 2 2 0 0 0 2 2 0 2 2 0 2 0 0 1 2 0 0 1 2 0 2 0 0 0 0 2 1 2 2 2 0 2
##    [297] 0 0 0 1 0 1 0 1 0 2 1 2 0 0 0 0 1 0 1 1 2 2 0 0 0 1 2 2 1 0 0 0 0 1 0 1 2
##    [334] 0 0 0 0 0 0 2 2 0 2 0 0 0 0 0 1 2 0 0 2 2 0 1 0 0 1 0 0 0 2 2 0 0 1 0 2 2
##    [371] 0 1 2 0 0 0 2 0 0 1 2 0 0 0 0 1 0 0 1 2 0 0 2 2 0 2 2 2 0 2 0 0 1 0 1 2 0
##    [408] 0 0 0 2 0 1 1 0 1 0 1 2 0 0 0 0 2 2 1 1 2 2 0 0 2 1 0 1 1 0 2 1 1 1 0 0 0
##    [445] 2 0 0 2 1 2 0 1 0 0 1 2 0 0 0 2 2 1 1 0 0 0 1 1 2 0 2 0 1 2 0 0 0 2 0 0 2
##    [482] 1 2 2 0 0 0 1 0 1 2 2 2 2 1 2 0 1 1 0 0 1 0 2 2 0 0 0 0 0 0 2 0 0 0 1 0 2
##    [519] 0 1 1 0 1 0 2 1 0 1 2 0 2 0 2 0 0 2 0 0 2 2 0 1 0 1 0 0 2 2 0 1 2 2 1 0 2
##    [556] 0 1 0 1 1 0 2 2 0 0 0 0 0 0 0 2 0 1 0 0 2 2 2 0 1 1 2 2 1 0 1 0 0 2 0 0 0 2
##    [593] 2 0 2 0 1 1 2 0 1 1 1 0 2 1 0 2 0 1 0 2 0 0 0 0 0 0 2 0 0 0 0 1 0 2 2 2 1
##    [630] 1 1 0 2 0 2 0 2 2 0 0 1 1 2 2 0 0 1 0 2 0 2 0 1 0 0 1 2 2 0 2 0 2 1 1 1 0
##    [667] 0 2 0 1 2 0 0 0 0 2 1 0 1 0 0 2 1 0 1 0 2 0 0 1 2 0 1 0 1 0 0 2 0 2 0 0 2
##    [704] 0 2 0 1 0 1 0 2 0 1 0 1 2 0 1 0 0 1 2 0 0 2 1 2 0 0 0 1 2 1 0 0 0 1 2 0 2
##    [741] 0 1 0 0 1 0 0 1 0 0 2 0 2 0 0 0 0 0 0 2 1 0 0 0 0 2 2 0 0 0 2 0 0 0 0 0 2
##    [778] 0 1 1 0 0 1 2 0 0 2 0 2 2 0 1 0 0 2 2 2 1 0 0 0 2 0 1 0 1 0 1 0 0 0 2 1 2
##    [815] 0 0 2 1 0 0 0 1 1 2 0 0 0 1 0 0 1 1 0 2 0 0 0 1 2 0 0 0 2 0 0 1 1 0 1 1 1
##    [852] 1 1 0 1 0 1 1 0 1 2 2 1 0 0 0 1 2 0 0 0 2 2 2 0 0 0 0 1 1 1 1 1 0 1 1 0 1
##    [889] 0 2 2 1 1 1 0 1 2 2 1 0 2 0 0 0 0 2 0 0 1 0 1 0 2 0 2 1 2 0 0 0 0 1 2 0 1
##    [926] 0 1 1 0 1 0 0 0 0 1 1 0 2 0 1 0 0 1 0 0 1 2 2 1 0 0 2 1 2 2 1 0 2 0 0 0 2
##    [963] 0 1 0 0 1 2 0 1 2 0 0 1 2 2 2 0 0 2 0 1 2 0 0 1 2 0 0 2 1 0 0 0 0 1 2 0 1
##   [1000] 0 0 2 1 2 0 2 1 2 1 2 0 1 2 0 0 1 1 1 0 2 0 0 0 1 0 2 2 0 2 2 2 1 2 2 0 1
##   [1037] 1 0 1 1 1 2 0 1 1 0 1 0 2 0 1 0 0 2 1 2 1 1 0 1 0 0 0 0 1 0 0 0 0 1 1 0 0
##   [1074] 2 1 2 0 0 0 2 0 0 2 0 2 2 0 0 0 2 0 1 0 0 1 0 2 1 0 0 0 1 0 2 2 1 0 1 1 2
##   [1111] 0 1 2 0 0 1 0 1 1 1 0 2 1 0 1 2 1 2 0 1 0 1 0 2 1 0 1 0 0 0 1 0 2 1 1 0 1
##   [1148] 2 0 0 0 2 1 0 2 1 2 0 1 2 0 1 2 1 0 0 1 0 0 0 2 0 0 0 1 1 0 2 2 1 2 2 0 1
##   [1185] 2 2 0 1 2 0 1 0 0 1 2 1 0 0 1 1 2 0 0 0 2 2 0 2 0 1 0 0 0 0 0 2 1 0 2 1 0
##   [1222] 1 1 2 0 1 0 0 0 2 0 1 0 2 0 0 2 2 0 1 2 1 0 0 1 0 1 0 2 0 2 0 0 0 0 0 2
##   [1259] 0 0 0 0 2 1 0 0 0 1 2 1 1 0 0 2 0 0 1 0 2 2 2 1 0 2 1 1 2 2 2 0 0 1 2 0 1
##   [1296] 0 0 0 1 0 0 1 0 0 2 2 0 0 1 1 0 0 0 1 2 2 0 2 0 0 0 1 1 0 1 2 1 2 2 2 0
##   [1333] 0 2 0 2 2 2 0 2 0 1 0 0 0 2 0 0 0 2 2 2 1 0 0 1 1 2 0 0 1 2 0 0 1 1 0 0 2
##   [1370] 2 0 0 2 0 1 0 0 1 1 0 0 0 0 0 0 0 0 1 1 0 1 0 2 0 2 0 0 0 0 2 1 0 0 0 2 0
##   [1407] 1 2 1 2 0 0 2 0 0 2 2 0 0 0 1 1 0 1 1 2 0 1 1 0 1 0 0 1 0 2 0 0 0 2 0 0 0
##   [1444] 1 0 2 1 2 0 0 0 0 2 0 1 1 1 0 1 0 0 0 1 1 0 1 2 0 0 0 0 1 1 0 1 0 0 1 1 1
##   [1481] 0 0 2 1 2 0 2 0 1 0 2 0 2 0 2 1 0 1 0 0 2 1 0 0 1 0 2 1 0 1 2 2 0 0 2 2 2
##   [1518] 0 2 0 2 1 0 0 0 1 1 0 1 0 2 0 1 0 0 2 1 2 0 0 0 0 2 0 0 1 0 1 0 0 0 2 0 1
##   [1555] 0 2 1 0 0 1 0 2 2 1 0 2 1 2 0 0 0 0 0 1 0 0 0 0 2 2 2 0 0 1 0 0 2 2 0 1 1
##   [1592] 2 0 0 1 2 1 0 1 1 2 0 0 0 0 0 0 0 2 0 2 0 2 2 0 2 2 0 0 1 0 0 1 0 0 1 1 1
##   [1629] 0 2 0 2 1 1 2 1 0 0 1 1 1 1 0 0 2 2 1 0 0 2 0 1 2 0 1 0 0 1 2 2 1 0 2 1 2
##   [1666] 0 0 0 0 0 1 0 0 2 0 2 0 0 1 2 2 0 2 0 0 2 0 1 0 0 1 0 0 1 1 2 1 0 0 2 1 0
##   [1703] 0 0 1 1 1 1 1 0 0 0 0 0 2 0 0 0 2 0 2 2 0 1 0 0 1 0 2 0 2 2 1 1 0 2 1 0 0
##   [1740] 2 0 0 1 1 2 2 0 2 0 0 0 0 0 1 0 0 2 1 0 0 0 2 0 0 2 0 2 0 0 2 1 2 0 0 0 1
##   [1777] 2 0 0 0 0 1 0 1 1 0 1 2 0 1 0 2 0 0 2 2 2 0 1 0 0 1 1 0 0 0 0 2 0 1 1 0 1
##   [1814] 0 1 2 1 0 0 0 0 0 0 0 0 0 0 1 0 0 2 1 0 2 0 0 1 1 2 0 2 0 0 2 2 0 0 0 0 0
##   [1851] 0 0 2 2 1 2 1 1 0 0 2 2 0 2 0 0 0 0 2 2 0 2 0 0 0 1 1 2 0 1 1 0 0 0 1 1 0
```

```

## [1888] 0 0 0 1 0 2 0 2 2 0 0 2 0 0 0 0 1 2 2 1 1 0 1 0 2 0 1 1 0 1 2 0 2 1 0 0 2
## [1925] 1 0 2 0 2 0 0 0 0 0 0 0 0 0 1 1 1 0 0 0 0 1 1 1 2 0 2 0 0 0 0 0 1 1 0 0 0 1
## [1962] 0 0 0 1 0 0 2 0 1 0 2 0 1 2 0 2 0 1 0 2 2 2 0 0 2 0 2 1 1 2 2 0 0 1 0 1 2
## [1999] 0 0 1 1 1 0 0 2 2 0 0 2 1 0 2 1 0 0 0 0 2 2 0 0 2 0 0 2 0 0 0 0 2 2 0 2 0
## [2036] 1 0 1 1 2 0 0 2 2 0 2 0 1 0 2 1 1 0 0 0 1 2 2 0 2 2 1 0 0 0 0 1 0 2 0 1 0
## [2073] 0 0 1 0 0 0 2 0 0 0 0 0 0 0 2 1 0 0 2 1 1 1 2 1 2 0 0 1 0 1 1 1 1 1 0 0 0 0
## [2110] 1 1 2 0 0 2 2 1 0 1 1 0 2 2 1 2 0 0 2 2 2 2 1 2 1 1 0 0 0 0 0 1 1 0 1 0 1
## [2147] 2 0 0 0 0 0 0 0 0 0 1 1 2 2 0 1 0 0 0 2 0 2 2 2 0 0 2 2 2 2 0 0 2 2 0 2 2 0
## [2184] 0 0 1 2 2 2 0 0 0 1 1 0 1 2 2 0 0 0 0 1 1 0 2 0 0 1 2 0 0 2 0 0 0 0 0 1 1
## [2221] 0 1 0 0 0 0 1 0 0 2 0 2 0 1 1 0 1 1 0 0 1 2 2 1 1 1 0 2 1 1 1 1 0 2 1 0 0
## [2258] 0 1 0 0 0 0 0 0 1 1 1 2 1 2 2 0 0 2 1 2 0 0 2 0 0 0 1 2 0 0 0 0 0 0 0 0 0
## [2295] 0 2 2 1 1 1 2 1 0 1 1 1 2 1 2 2 0 2 0 0 0 2 0 0 0 1 2 0 0 0 0 0 1 0 0 0 1
## [2332] 1 2 2 0 2 2 1 2 2 0 0 2 1 1 0 0 1 1 2 0 0 2 0 0 0 0 0 0 2 0 0 2 0 1 0 2 0
## [2369] 1 1 0 2 0 0 1 2 0 0 0 0 0 0 1 0 2 0 0 0 2 1 0 1 0 0 0 2 0 0 2 0 0 1 0 1 1
## [2406] 0 2 0 2 1 2 0 1 0 0 2 0 1 0 1 0 0 1 0 0 2 1 0 0 2 2 2 0 1 0 1 2 2 2 2 1 2
## [2443] 0 0 1 1 0 0 0 1 1 2 2 2 0 2 2 0 0 1 1 2 0 2 0 1 1 1 0 2 1 1 1 0 0 2 1 1 0
## [2480] 2 1 0 2 0 2 2 1 2 2 1 1 0 0 1 2 0 2 1 1 0

```

```
R = matrix(vec, nrow = 50, ncol = 50)
```

```
R
```

```

##      [,1] [,2] [,3] [,4] [,5] [,6] [,7] [,8] [,9] [,10] [,11] [,12] [,13]
## [1,]    1    0    0    1    2    0    0    0    0    0    0    2    1
## [2,]    1    0    2    2    0    2    1    0    0    1    1    2    1
## [3,]    2    1    0    2    1    1    0    2    1    0    0    1    1
## [4,]    0    2    0    2    1    1    1    2    0    0    2    0    0
## [5,]    1    0    2    2    0    0    0    0    1    1    2    2    2
## [6,]    1    0    0    2    0    0    2    1    2    2    0    0    1
## [7,]    1    0    0    0    1    0    1    0    0    0    0    1    0
## [8,]    0    0    0    2    0    2    2    0    0    0    0    0    2
## [9,]    0    0    0    0    0    2    0    1    0    0    0    1    0
## [10,]   2    0    1    1    1    0    0    0    0    2    0    1    1
## [11,]   0    0    0    2    2    0    0    0    2    2    0    0    0
## [12,]   1    2    0    0    0    0    0    0    0    1    2    2    2
## [13,]   0    2    0    0    2    0    1    2    1    1    0    2    0
## [14,]   0    2    0    0    0    2    0    2    1    0    0    0    0
## [15,]   0    0    1    1    1    2    1    0    0    0    0    0    0
## [16,]   2    0    1    2    0    0    1    0    1    0    1    0    0
## [17,]   2    0    2    2    1    0    2    1    0    1    0    0    0
## [18,]   2    0    1    0    0    0    2    0    1    1    2    0    0
## [19,]   1    2    0    1    1    2    0    2    2    2    0    0    2
## [20,]   0    0    2    0    0    2    0    2    0    0    1    2    0
## [21,]   1    0    0    0    1    0    0    0    0    2    1    0    0
## [22,]   2    0    0    0    2    2    1    1    0    0    0    1    0
## [23,]   0    0    2    2    0    2    2    2    0    1    1    0    0
## [24,]   1    1    0    0    0    0    2    0    2    2    0    0    1
## [25,]   1    0    1    1    0    2    1    0    2    0    2    2    0
## [26,]   1    0    1    0    2    0    0    0    1    0    1    2    2
## [27,]   0    0    2    0    0    0    0    2    1    0    0    2    2
## [28,]   0    0    2    0    0    1    0    0    2    2    1    0    2
## [29,]   2    2    0    0    2    2    0    0    2    0    2    1    1
## [30,]   1    0    0    0    2    0    1    1    0    0    0    1    1
## [31,]   2    2    1    2    0    0    0    2    0    2    2    2    1
## [32,]   2    0    1    0    0    1    1    0    2    1    0    2    0
## [33,]   0    0    1    0    1    2    2    0    1    2    2    1    2

```

## [34,]	0	2	2	2	0	0	0	0	0	2	0	0	0
## [35,]	0	1	0	1	1	2	0	0	1	0	0	1	2
## [36,]	2	0	0	0	0	0	0	1	1	0	2	0	0
## [37,]	0	0	0	1	0	0	0	0	0	0	0	0	2
## [38,]	0	0	0	0	0	0	0	0	2	1	0	2	2
## [39,]	1	0	1	2	0	0	0	1	1	0	2	0	0
## [40,]	0	0	1	0	2	2	2	2	1	1	2	0	0
## [41,]	0	0	0	1	1	1	2	0	1	2	0	0	1
## [42,]	1	0	1	2	1	2	0	0	0	2	1	2	1
## [43,]	0	0	1	2	2	2	2	2	0	2	0	2	2
## [44,]	0	2	2	1	0	2	0	2	0	2	1	0	2
## [45,]	0	0	0	2	1	0	0	0	2	1	0	2	0
## [46,]	1	0	1	0	2	2	0	2	0	2	0	0	0
## [47,]	2	0	2	0	0	0	0	2	0	0	2	1	1
## [48,]	0	0	0	1	0	0	0	2	2	1	2	1	0
## [49,]	1	0	1	0	2	0	1	0	1	1	0	2	2
## [50,]	0	2	2	0	0	1	2	2	2	0	1	0	0
##	[,14]	[,15]	[,16]	[,17]	[,18]	[,19]	[,20]	[,21]	[,22]	[,23]	[,24]	[,25]	
## [1,]	2	0	2	0	1	2	0	0	1	0	0	0	2
## [2,]	0	0	0	2	1	0	2	2	0	1	2	0	
## [3,]	1	2	2	0	1	0	1	1	0	0	1	0	
## [4,]	0	0	0	1	0	0	2	2	2	2	0	0	
## [5,]	0	2	0	0	1	0	2	0	1	2	2	2	
## [6,]	1	0	0	1	0	2	1	2	2	1	1	2	
## [7,]	2	1	0	0	1	0	0	1	1	0	2	0	
## [8,]	2	0	0	1	1	0	2	2	1	1	0	2	
## [9,]	0	1	0	0	0	1	0	1	0	1	1	0	
## [10,]	2	0	2	0	1	0	0	2	1	2	2	1	
## [11,]	0	2	1	0	2	1	0	0	0	0	0	0	
## [12,]	2	0	0	2	2	0	2	1	0	1	1	0	
## [13,]	1	1	0	1	1	2	0	2	0	2	2	0	
## [14,]	1	0	0	2	0	0	1	0	0	0	1	0	
## [15,]	1	1	0	0	0	2	0	0	1	0	0	0	
## [16,]	0	2	2	0	0	1	0	1	0	1	0	2	
## [17,]	0	0	2	2	1	2	1	1	0	0	1	1	
## [18,]	2	1	0	1	2	0	2	1	0	1	0	0	
## [19,]	0	0	0	0	0	0	0	0	0	1	0	2	
## [20,]	1	0	0	0	0	0	1	2	1	1	0	1	
## [21,]	2	1	2	0	0	0	2	0	1	0	2	0	
## [22,]	0	2	0	1	2	1	0	0	0	2	0	1	
## [23,]	0	0	0	1	2	2	0	0	0	1	0	1	
## [24,]	0	0	0	2	2	0	1	1	2	0	0	2	
## [25,]	0	2	0	0	0	1	2	0	1	1	1	0	
## [26,]	2	1	0	0	0	0	2	2	2	2	1	1	
## [27,]	1	2	2	0	0	1	2	2	0	1	0	0	
## [28,]	0	0	0	1	0	1	0	0	0	2	2	0	
## [29,]	1	0	1	0	1	0	0	2	0	0	2	0	
## [30,]	0	0	1	0	1	1	2	2	2	1	1	2	
## [31,]	0	1	0	1	1	0	0	2	0	0	2	0	
## [32,]	2	2	0	1	1	0	1	1	0	1	2	1	
## [33,]	1	1	1	0	1	0	2	2	2	0	0	0	
## [34,]	0	0	2	2	0	0	0	2	0	2	1	2	
## [35,]	1	0	0	0	1	1	0	0	2	1	2	0	
## [36,]	0	0	0	0	1	1	1	1	2	0	2	0	

## [37,]	2	1	2	0	0	0	2	1	0	1	0	2
## [38,]	0	2	0	1	1	2	0	0	0	0	1	2
## [39,]	0	0	2	2	0	0	0	1	0	0	2	0
## [40,]	1	2	2	0	2	1	2	1	2	0	0	1
## [41,]	2	0	0	0	2	0	1	1	0	1	1	2
## [42,]	0	1	1	0	1	0	0	2	1	0	0	1
## [43,]	1	0	0	2	1	1	0	0	0	2	0	0
## [44,]	0	0	0	0	1	0	0	1	0	1	1	0
## [45,]	1	1	2	0	0	0	0	1	1	1	2	1
## [46,]	0	0	2	1	1	1	1	0	0	0	1	0
## [47,]	0	0	2	1	2	2	2	1	2	1	0	1
## [48,]	2	1	1	0	2	2	0	0	1	2	0	0
## [49,]	0	0	0	1	1	1	1	2	0	0	1	2
## [50,]	2	0	0	1	0	0	0	0	0	0	1	0
##	[,26]	[,27]	[,28]	[,29]	[,30]	[,31]	[,32]	[,33]	[,34]	[,35]	[,36]	[,37]
## [1,]	2	0	2	1	0	2	0	2	0	1	0	0
## [2,]	0	1	2	0	0	1	2	0	1	0	0	1
## [3,]	0	0	1	0	2	0	0	0	2	0	0	1
## [4,]	0	0	0	0	0	0	1	0	0	0	1	0
## [5,]	0	2	0	2	1	1	0	0	1	1	0	0
## [6,]	0	2	1	0	1	0	2	0	0	1	0	0
## [7,]	0	0	1	1	1	2	1	0	0	1	2	0
## [8,]	2	0	2	2	0	1	0	0	1	1	1	2
## [9,]	0	1	0	1	1	0	0	2	2	1	0	0
## [10,]	0	1	0	2	0	1	1	0	2	0	0	1
## [11,]	0	0	1	0	0	2	0	2	1	0	0	1
## [12,]	0	0	2	0	0	2	2	0	0	0	2	0
## [13,]	2	0	0	2	1	0	2	2	2	0	0	1
## [14,]	1	1	0	0	1	0	1	2	1	0	0	0
## [15,]	0	2	1	0	0	2	0	0	2	2	2	1
## [16,]	0	2	1	2	1	2	2	2	0	0	0	2
## [17,]	0	0	0	2	2	2	1	2	0	0	2	1
## [18,]	1	2	0	0	0	0	2	0	0	0	0	0
## [19,]	2	0	2	0	0	2	0	0	0	2	0	0
## [20,]	1	0	2	0	0	0	0	1	0	0	2	0
## [21,]	1	0	0	1	0	2	0	0	1	2	1	0
## [22,]	0	1	0	1	1	1	0	0	0	2	2	0
## [23,]	0	1	2	0	1	0	0	1	0	0	0	0
## [24,]	2	0	0	1	0	0	1	0	2	1	0	0
## [25,]	0	1	1	1	1	0	0	0	0	0	0	0
## [26,]	0	2	0	2	0	1	0	1	2	0	1	0
## [27,]	1	1	0	0	0	1	0	1	0	1	2	1
## [28,]	0	2	1	1	1	0	0	1	0	0	0	0
## [29,]	2	2	1	1	1	1	2	0	1	2	0	0
## [30,]	2	2	0	0	1	0	2	2	2	0	0	2
## [31,]	2	2	0	1	0	2	2	0	2	2	0	1
## [32,]	1	0	0	0	0	0	0	2	0	2	1	0
## [33,]	0	0	0	0	2	1	0	1	2	1	0	2
## [34,]	2	2	0	1	1	0	1	1	0	1	1	0
## [35,]	1	0	0	0	2	0	0	2	0	0	1	0
## [36,]	1	2	0	2	0	2	0	1	2	2	0	1
## [37,]	2	2	0	0	2	1	2	0	0	1	1	1
## [38,]	2	2	1	0	0	2	2	0	1	0	2	2
## [39,]	2	0	1	0	1	0	0	1	0	0	0	0

## [40,]	0	2	0	2	0	0	1	1	0	2	1	2
## [41,]	0	0	1	0	2	0	1	1	1	0	0	0
## [42,]	1	1	0	0	0	0	2	1	0	0	2	0
## [43,]	2	0	2	0	2	2	0	0	0	1	0	2
## [44,]	0	0	0	1	0	0	0	0	1	1	0	2
## [45,]	1	0	2	0	2	0	1	2	1	2	2	0
## [46,]	0	2	0	2	1	1	2	2	2	2	2	0
## [47,]	0	0	0	1	0	0	1	1	1	0	2	0
## [48,]	0	0	0	2	1	1	0	0	0	2	0	0
## [49,]	1	0	0	0	0	0	1	0	0	0	1	0
## [50,]	0	2	2	0	0	0	1	2	2	0	0	0
##	[,38]	[,39]	[,40]	[,41]	[,42]	[,43]	[,44]	[,45]	[,46]	[,47]	[,48]	[,49]
## [1,]	0	0	0	1	1	1	0	0	1	2	0	0
## [2,]	0	0	0	1	1	1	0	0	1	1	0	1
## [3,]	2	0	0	1	0	1	0	1	0	0	2	0
## [4,]	2	1	0	0	0	1	0	1	2	1	0	1
## [5,]	1	2	0	0	0	1	0	0	1	1	0	1
## [6,]	2	2	1	2	1	0	1	2	0	1	0	0
## [7,]	1	1	1	2	2	0	1	0	0	2	0	2
## [8,]	1	1	0	0	2	0	2	0	0	1	0	0
## [9,]	0	0	0	0	0	0	2	1	1	2	0	2
## [10,]	0	1	0	2	2	1	0	2	0	2	2	1
## [11,]	2	0	1	1	2	1	1	0	0	0	0	2
## [12,]	2	2	0	0	1	2	0	0	0	2	0	0
## [13,]	0	0	0	2	0	0	0	2	0	0	2	1
## [14,]	2	1	0	1	0	0	0	0	0	0	0	0
## [15,]	0	1	1	0	0	2	2	0	0	0	1	0
## [16,]	0	0	0	0	0	2	0	0	1	2	0	2
## [17,]	0	1	0	0	1	1	2	0	1	0	2	0
## [18,]	0	2	2	0	0	0	2	0	1	0	0	1
## [19,]	2	0	0	2	2	1	2	1	2	0	1	0
## [20,]	2	2	1	2	0	1	0	1	1	1	1	1
## [21,]	0	1	0	0	1	0	0	0	2	2	0	0
## [22,]	2	0	2	0	0	2	2	1	2	0	2	0
## [23,]	0	0	0	2	0	2	2	0	0	0	0	1
## [24,]	0	2	1	0	0	1	2	0	0	0	0	0
## [25,]	0	1	2	0	1	2	2	0	2	0	1	0
## [26,]	1	0	0	2	0	0	0	0	1	0	2	2
## [27,]	1	2	2	0	0	0	0	1	2	1	0	1
## [28,]	2	0	0	0	0	2	2	0	0	0	0	0
## [29,]	0	2	1	0	2	2	2	0	0	0	0	0
## [30,]	1	0	0	0	0	2	0	2	2	0	0	2
## [31,]	1	0	2	2	0	2	2	0	0	1	0	2
## [32,]	0	0	2	2	0	1	2	2	0	1	0	2
## [33,]	0	0	2	0	0	2	0	0	0	2	1	0
## [34,]	0	0	0	2	0	1	0	1	1	2	0	1
## [35,]	1	0	0	0	0	1	0	1	2	0	2	0
## [36,]	1	0	2	1	2	0	1	0	0	2	0	1
## [37,]	0	0	0	0	1	0	2	1	0	2	0	2
## [38,]	0	1	2	1	0	0	2	1	0	1	0	2
## [39,]	0	1	1	1	0	0	2	0	0	2	2	2
## [40,]	0	1	1	2	2	0	0	0	0	2	1	2
## [41,]	1	0	2	0	1	1	0	1	0	0	0	1
## [42,]	0	0	2	0	1	1	0	2	0	0	1	2

## [43,]	2	0	0	2	1	0	1	2	0	2	0	0
## [44,]	0	0	0	2	2	1	1	1	0	1	0	0
## [45,]	2	1	1	0	1	0	0	1	0	1	0	1
## [46,]	2	1	0	2	2	1	1	1	2	0	2	1
## [47,]	0	1	1	0	0	2	2	0	2	0	0	0
## [48,]	0	2	2	1	0	0	2	2	1	1	0	0
## [49,]	2	0	0	0	1	0	0	1	1	1	2	0
## [50,]	0	2	0	2	0	0	0	1	1	2	0	1
##	[,50]											
## [1,]	1											
## [2,]	2											
## [3,]	2											
## [4,]	2											
## [5,]	0											
## [6,]	2											
## [7,]	2											
## [8,]	0											
## [9,]	0											
## [10,]	1											
## [11,]	1											
## [12,]	2											
## [13,]	0											
## [14,]	2											
## [15,]	0											
## [16,]	1											
## [17,]	1											
## [18,]	1											
## [19,]	0											
## [20,]	2											
## [21,]	1											
## [22,]	1											
## [23,]	1											
## [24,]	0											
## [25,]	0											
## [26,]	2											
## [27,]	1											
## [28,]	1											
## [29,]	0											
## [30,]	2											
## [31,]	1											
## [32,]	0											
## [33,]	2											
## [34,]	0											
## [35,]	2											
## [36,]	2											
## [37,]	1											
## [38,]	2											
## [39,]	2											
## [40,]	1											
## [41,]	1											
## [42,]	0											
## [43,]	0											
## [44,]	1											
## [45,]	2											

```
## [46,] 0
## [47,] 2
## [48,] 1
## [49,] 1
## [50,] 0
```

```
table(c(R)) #table lists out the unique value of the matrix and inputing a matrix into c makes the matr
```

```
##
##      0      1      2
## 1250  625  625
```

- Randomly punch holes (i.e. NA) values in this matrix so that an each entry is missing with probability 30%.

```
R[runif(2500)<0.3] = NA
R
```

```
##      [,1] [,2] [,3] [,4] [,5] [,6] [,7] [,8] [,9] [,10] [,11] [,12] [,13]
## [1,] 1    0    0    1    NA    0    NA    0    0    NA    0    2    1
## [2,] NA    0    NA    NA    NA    2    1    NA    0    NA    1    2    1
## [3,] NA    1    0    2    1    NA    NA    NA    1    0    0    1    NA
## [4,] 0    2    0    NA    1    1    1    2    0    0    2    0    0
## [5,] 1    0    2    2    NA    NA    NA    0    NA    1    2    2    2
## [6,] 1    0    0    2    0    NA    2    1    NA    2    0    0    NA
## [7,] 1    0    NA    NA    1    0    NA    0    0    NA    NA    1    NA
## [8,] NA    0    0    2    0    2    2    0    0    0    0    NA    2
## [9,] NA    0    0    0    0    2    0    1    0    NA    NA    1    0
## [10,] 2    0    NA    NA    1    NA    0    0    NA    2    0    1    1
## [11,] 0    NA    0    2    2    0    NA    0    NA    NA    0    0    0
## [12,] 1    NA    0    0    0    0    0    NA    0    NA    2    2    2
## [13,] 0    2    0    NA    2    0    1    NA    1    1    0    2    0
## [14,] 0    2    0    0    0    2    NA    NA    NA    0    NA    0    0
## [15,] NA    0    1    1    NA    NA    1    0    NA    NA    0    0    NA
## [16,] 2    0    1    NA    0    NA    1    0    1    0    NA    0    0
## [17,] 2    0    2    NA    1    0    2    1    NA    1    0    0    0
## [18,] NA    0    1    0    0    0    2    NA    1    1    2    0    0
## [19,] 1    2    0    1    NA    NA    0    2    NA    NA    0    0    2
## [20,] NA    NA    2    0    0    NA    0    2    0    0    1    NA    0
## [21,] 1    0    0    0    1    NA    0    0    0    2    NA    0    0
## [22,] 2    NA    NA    0    NA    2    1    NA    0    NA    0    1    0
## [23,] 0    0    2    NA    0    NA    2    2    0    NA    1    0    0
## [24,] 1    1    0    0    0    0    2    0    NA    2    NA    0    1
## [25,] 1    0    1    1    NA    2    1    0    2    0    2    2    NA
## [26,] NA    0    1    0    2    0    0    0    1    0    1    2    NA
## [27,] 0    0    NA    0    0    0    0    2    NA    NA    NA    NA    NA
## [28,] 0    0    2    0    NA    1    NA    NA    2    2    1    NA    2
## [29,] 2    2    NA    0    NA    2    0    NA    2    NA    NA    1    1
## [30,] NA    0    0    NA    2    NA    1    1    NA    0    0    1    1
## [31,] 2    NA    1    NA    0    0    0    2    0    2    2    2    NA
## [32,] 2    0    1    0    0    NA    1    0    2    NA    0    2    0
## [33,] 0    NA    1    NA    1    2    2    NA    NA    NA    2    NA    2
## [34,] NA    2    2    NA    NA    NA    0    NA    0    2    0    0    0
## [35,] 0    NA    NA    1    1    NA    NA    0    1    0    0    1    2
## [36,] NA    0    0    0    0    0    0    NA    NA    1    NA    2    0
## [37,] 0    0    NA    1    0    0    0    NA    0    0    NA    NA    NA
```

## [38,]	0	0	0	NA	0	NA	0	NA	2	1	NA	2	2
## [39,]	NA	NA	1	2	0	0	0	1	1	0	NA	0	0
## [40,]	NA	0	NA	NA	NA	NA	2	NA	1	1	2	0	0
## [41,]	NA	NA	NA	1	1	NA	NA	NA	1	NA	0	NA	NA
## [42,]	1	0	NA	NA	1	NA	0	0	0	2	NA	NA	NA
## [43,]	0	0	1	2	NA	2	2	2	0	2	NA	2	2
## [44,]	0	2	2	1	NA	NA	NA	2	0	NA	NA	NA	2
## [45,]	0	0	0	2	1	0	0	0	2	NA	NA	2	0
## [46,]	1	0	1	0	NA	2	0	2	NA	2	0	NA	0
## [47,]	2	NA	2	NA	0	0	NA	2	0	0	2	1	1
## [48,]	0	0	0	NA	0	0	0	NA	2	NA	2	1	0
## [49,]	1	NA	1	NA	2	0	1	0	1	1	0	2	2
## [50,]	0	2	NA	0	0	1	NA	NA	NA	NA	1	0	0
##	[,14]	[,15]	[,16]	[,17]	[,18]	[,19]	[,20]	[,21]	[,22]	[,23]	[,24]	[,25]	
## [1,]	2	0	2	0	1	2	0	0	1	NA	NA	2	
## [2,]	NA	0	0	NA	NA	NA	2	2	0	1	2	0	
## [3,]	1	2	2	NA	NA	0	1	1	0	0	NA	NA	
## [4,]	0	NA	0	1	NA	NA	2	2	NA	NA	0	NA	
## [5,]	NA	2	0	0	NA	0	NA	0	NA	2	2	2	
## [6,]	1	0	NA	NA	0	2	1	2	NA	1	1	NA	
## [7,]	2	1	0	0	1	NA	0	1	1	NA	2	NA	
## [8,]	2	0	0	1	1	NA	NA	2	1	1	0	2	
## [9,]	NA	1	0	0	NA	NA	NA	NA	0	NA	1	0	
## [10,]	2	0	2	NA	NA	0	0	2	1	2	NA	NA	
## [11,]	NA	2	1	0	NA	1	0	0	0	0	0	0	
## [12,]	2	0	NA	2	2	0	NA	1	0	NA	1	0	
## [13,]	1	NA	NA	1	NA	2	0	2	NA	2	2	0	
## [14,]	1	NA	0	2	NA	0	1	0	NA	0	NA	0	
## [15,]	NA	1	0	NA	0	2	0	0	1	0	0	0	
## [16,]	0	2	NA	0	0	1	0	1	0	1	0	2	
## [17,]	NA	0	2	NA	1	NA	1	1	0	0	1	NA	
## [18,]	2	1	0	1	2	NA	NA	1	0	1	0	0	
## [19,]	NA	0	0	NA	NA	NA	0	NA	0	1	0	2	
## [20,]	1	0	NA	0	0	0	NA	2	1	1	0	NA	
## [21,]	2	1	2	0	NA	NA	2	0	NA	0	2	0	
## [22,]	0	2	0	NA	NA	1	0	NA	0	NA	0	1	
## [23,]	0	0	NA	1	2	NA	0	0	0	1	0	NA	
## [24,]	0	0	0	2	2	0	NA	1	NA	0	0	2	
## [25,]	0	2	0	0	0	1	NA	NA	NA	NA	1	0	
## [26,]	2	1	NA	0	0	NA	2	NA	2	2	1	1	
## [27,]	NA	2	2	NA	0	NA	2	NA	0	NA	0	0	
## [28,]	0	0	NA	1	0	1	0	0	NA	2	2	0	
## [29,]	1	0	1	0	1	0	0	2	NA	0	2	NA	
## [30,]	0	0	NA	0	1	1	2	2	NA	1	NA	2	
## [31,]	0	1	0	NA	1	NA	0	NA	0	0	NA	0	
## [32,]	2	2	NA	1	1	0	1	1	NA	NA	NA	1	
## [33,]	1	1	1	0	1	NA	2	2	NA	0	0	0	
## [34,]	0	NA	2	2	0	NA	NA	2	0	2	1	2	
## [35,]	1	NA	NA	NA	1	1	0	0	2	NA	NA	0	
## [36,]	0	0	NA	0	1	1	NA	1	2	NA	NA	0	
## [37,]	2	1	NA	NA	0	NA	NA	NA	0	NA	0	2	
## [38,]	0	2	0	1	1	2	NA	0	0	0	1	NA	
## [39,]	0	0	2	2	0	0	NA	1	NA	0	2	0	
## [40,]	1	2	2	0	NA	NA	2	1	NA	NA	0	NA	

## [41,]	2	0	NA	NA	NA	0	1	1	0	1	1	NA
## [42,]	0	1	1	NA	1	0	0	2	NA	NA	NA	NA
## [43,]	1	NA	0	2	1	1	0	NA	0	NA	0	0
## [44,]	0	0	0	0	NA	NA	0	1	0	1	1	0
## [45,]	1	NA	2	0	NA	0	0	NA	1	NA	2	NA
## [46,]	0	NA	2	NA	1	1	1	0	0	0	1	0
## [47,]	0	0	2	1	2	2	NA	1	NA	NA	0	1
## [48,]	NA	1	1	0	2	2	0	0	1	2	0	NA
## [49,]	0	0	0	1	1	1	NA	NA	0	0	1	NA
## [50,]	2	0	0	1	0	NA	0	0	0	0	NA	0
##	[,26]	[,27]	[,28]	[,29]	[,30]	[,31]	[,32]	[,33]	[,34]	[,35]	[,36]	[,37]
## [1,]	NA	0	2	1	0	2	0	2	0	1	0	NA
## [2,]	0	1	2	0	NA	NA	2	NA	NA	0	0	1
## [3,]	NA	0	1	0	2	NA	0	0	2	0	0	NA
## [4,]	0	0	0	0	0	NA	1	NA	0	0	1	0
## [5,]	0	NA	0	2	1	1	NA	0	NA	1	0	NA
## [6,]	0	2	1	0	1	NA	2	NA	NA	1	0	0
## [7,]	NA	0	NA	NA	1	2	1	NA	0	NA	NA	0
## [8,]	2	NA	2	2	0	1	NA	NA	NA	1	1	NA
## [9,]	0	1	NA	1	1	0	0	NA	NA	NA	0	0
## [10,]	NA	1	0	2	0	1	1	0	NA	0	0	NA
## [11,]	0	NA	1	0	0	2	0	2	NA	0	NA	1
## [12,]	0	0	2	0	0	2	NA	0	NA	NA	NA	0
## [13,]	2	0	0	NA	1	0	2	NA	2	0	0	1
## [14,]	NA	1	0	NA	1	0	1	NA	1	0	0	0
## [15,]	0	NA	NA	0	0	NA	0	0	2	2	2	NA
## [16,]	0	2	1	2	NA	NA	NA	2	0	0	NA	2
## [17,]	NA	0	NA	2	2	2	1	NA	0	NA	NA	1
## [18,]	1	NA	0	NA	0	0	NA	0	0	0	0	NA
## [19,]	NA	0	2	0	0	2	0	NA	0	2	0	0
## [20,]	NA	0	NA	0	0	0	0	1	0	NA	NA	0
## [21,]	1	0	0	NA	NA	2	0	0	1	NA	NA	NA
## [22,]	0	1	0	1	1	1	0	0	0	2	NA	0
## [23,]	0	1	2	0	1	0	0	1	0	NA	NA	NA
## [24,]	2	0	0	1	0	0	1	0	2	NA	NA	0
## [25,]	NA	1	1	1	1	0	NA	NA	NA	NA	NA	0
## [26,]	0	2	0	NA	0	1	0	NA	NA	NA	1	0
## [27,]	1	1	NA	0	0	NA	0	1	0	NA	2	1
## [28,]	NA	2	1	NA	1	0	0	1	NA	NA	NA	0
## [29,]	2	2	1	NA	NA	1	NA	NA	1	2	0	NA
## [30,]	2	NA	0	0	1	0	2	NA	2	0	NA	2
## [31,]	NA	NA	0	1	NA	NA	2	NA	NA	2	NA	NA
## [32,]	1	0	0	NA	NA	0	NA	2	0	2	NA	0
## [33,]	0	NA	0	0	2	1	0	1	2	1	0	2
## [34,]	2	NA	0	NA	NA	0	NA	NA	0	1	1	0
## [35,]	NA	0	0	NA	2	0	0	2	0	0	1	NA
## [36,]	NA	2	0	NA	0	NA	0	NA	NA	NA	0	NA
## [37,]	2	NA	0	NA	2	1	2	NA	0	1	1	1
## [38,]	2	2	NA	0	0	2	2	0	1	NA	NA	2
## [39,]	2	0	NA	NA	1	0	0	1	NA	0	0	0
## [40,]	0	2	0	2	0	0	NA	NA	0	2	1	2
## [41,]	0	0	1	0	2	0	1	1	NA	0	0	NA
## [42,]	1	NA	0	0	0	0	NA	1	NA	0	2	NA
## [43,]	2	0	2	NA	NA	2	NA	0	0	1	NA	2

##	[44,]	0	NA	0	1	0	NA	NA	0	1	1	0	2
##	[45,]	NA	0	2	NA	NA	0	1	2	NA	2	NA	0
##	[46,]	0	2	0	2	1	1	2	2	2	2	2	NA
##	[47,]	0	NA	0	1	0	0	1	1	1	NA	2	0
##	[48,]	NA	0	0	NA	NA	1	NA	0	0	NA	0	NA
##	[49,]	1	NA	0	0	0	0	1	0	0	NA	NA	NA
##	[50,]	0	2	2	0	NA	NA	1	NA	2	0	NA	0
##		[,38]	[,39]	[,40]	[,41]	[,42]	[,43]	[,44]	[,45]	[,46]	[,47]	[,48]	[,49]
##	[1,]	0	0	NA	NA	1	1	NA	NA	1	2	0	0
##	[2,]	0	0	0	1	1	1	0	0	NA	1	0	1
##	[3,]	2	0	NA	1	NA	1	0	1	NA	0	2	0
##	[4,]	NA	1	0	NA	0	1	0	1	2	1	0	NA
##	[5,]	NA	NA	0	0	0	1	0	0	1	1	0	1
##	[6,]	2	2	1	2	1	0	NA	2	0	1	0	0
##	[7,]	NA	NA	1	2	NA	0	1	0	NA	2	0	NA
##	[8,]	1	1	0	0	2	NA	2	0	0	1	0	0
##	[9,]	0	0	0	NA	0	0	2	NA	1	NA	0	2
##	[10,]	0	NA	NA	2	2	1	0	2	0	2	2	1
##	[11,]	2	0	NA	1	NA	1	1	0	0	0	NA	2
##	[12,]	2	NA	0	0	1	NA	NA	NA	NA	2	0	NA
##	[13,]	0	0	0	2	0	NA	NA	2	0	0	NA	1
##	[14,]	2	1	0	1	0	0	NA	0	0	NA	0	0
##	[15,]	0	1	1	NA	NA	NA	2	NA	0	0	1	0
##	[16,]	NA	0	0	NA	NA	2	0	0	NA	2	NA	2
##	[17,]	0	1	NA	0	1	NA	2	0	NA	0	2	NA
##	[18,]	0	2	NA	NA	NA	NA	2	0	1	0	0	NA
##	[19,]	2	0	0	2	2	1	2	1	2	0	1	NA
##	[20,]	2	2	1	2	NA	1	NA	1	NA	1	1	1
##	[21,]	0	NA	NA	NA	1	0	0	0	NA	2	NA	0
##	[22,]	NA	0	2	NA	NA	2	2	1	2	NA	2	0
##	[23,]	0	NA	0	NA	0	2	NA	0	0	0	NA	NA
##	[24,]	0	NA	1	0	0	NA	2	0	NA	0	0	0
##	[25,]	0	NA	2	0	1	2	2	0	2	0	1	0
##	[26,]	1	0	0	2	0	0	0	NA	1	0	NA	2
##	[27,]	1	NA	2	NA	0	0	0	1	2	1	0	NA
##	[28,]	2	0	0	0	0	NA	2	NA	0	0	NA	0
##	[29,]	0	2	1	NA	2	NA	2	0	NA	0	0	0
##	[30,]	1	0	NA	0	0	2	0	2	2	0	NA	NA
##	[31,]	1	NA	2	2	0	2	2	0	0	NA	0	2
##	[32,]	NA	0	2	2	0	1	2	NA	0	1	NA	2
##	[33,]	NA	NA	2	NA	0	2	0	NA	0	2	NA	0
##	[34,]	NA	0	0	2	NA	NA	0	1	1	NA	0	1
##	[35,]	NA	0	0	0	0	1	0	1	NA	0	2	0
##	[36,]	1	NA	2	1	2	0	1	NA	0	2	0	1
##	[37,]	NA	0	0	0	1	NA	2	1	NA	2	0	2
##	[38,]	0	NA	2	1	NA	NA	NA	NA	NA	1	NA	2
##	[39,]	0	1	NA	1	NA	0	NA	NA	0	2	2	2
##	[40,]	0	1	1	2	2	NA	0	0	0	2	1	NA
##	[41,]	1	0	2	NA	NA	1	0	NA	0	0	0	1
##	[42,]	0	0	2	0	1	1	0	NA	0	NA	1	2
##	[43,]	NA	NA	0	2	NA	0	1	2	NA	NA	0	NA
##	[44,]	NA	0	0	NA	2	1	1	1	0	1	0	0
##	[45,]	NA	1	1	NA	NA	NA	0	1	0	1	0	1
##	[46,]	2	NA	0	2	2	NA	NA	NA	2	NA	2	NA

##	[47,]	0	NA	1	NA	0	2	2	0	2	0	0	0
##	[48,]	0	NA	2	1	NA	0	2	2	1	NA	0	NA
##	[49,]	NA	0	NA	NA	1	NA	0	1	NA	NA	2	NA
##	[50,]	NA	2	NA	2	NA	NA	0	1	1	2	0	1
##	[,50]												
##	[1,]	1											
##	[2,]	NA											
##	[3,]	2											
##	[4,]	NA											
##	[5,]	0											
##	[6,]	2											
##	[7,]	2											
##	[8,]	0											
##	[9,]	0											
##	[10,]	NA											
##	[11,]	1											
##	[12,]	NA											
##	[13,]	0											
##	[14,]	2											
##	[15,]	NA											
##	[16,]	NA											
##	[17,]	NA											
##	[18,]	NA											
##	[19,]	0											
##	[20,]	2											
##	[21,]	1											
##	[22,]	1											
##	[23,]	NA											
##	[24,]	0											
##	[25,]	0											
##	[26,]	NA											
##	[27,]	1											
##	[28,]	NA											
##	[29,]	0											
##	[30,]	NA											
##	[31,]	1											
##	[32,]	0											
##	[33,]	2											
##	[34,]	0											
##	[35,]	2											
##	[36,]	NA											
##	[37,]	1											
##	[38,]	NA											
##	[39,]	2											
##	[40,]	1											
##	[41,]	1											
##	[42,]	NA											
##	[43,]	0											
##	[44,]	1											
##	[45,]	2											
##	[46,]	NA											
##	[47,]	2											
##	[48,]	NA											
##	[49,]	NA											

```
## [50,]      0
```

- Sort the rows in matrix R by the largest row sum to lowest. Be careful about the NA's!

```
R[order(rowSums(R, na.rm = TRUE), decreasing = TRUE),]
```

```
##      [,1] [,2] [,3] [,4] [,5] [,6] [,7] [,8] [,9] [,10] [,11] [,12] [,13]
## [1,]    1    0    1    0   NA    2    0    2   NA    2    0   NA    0
## [2,]    1    0    0    2    0   NA    2    1   NA    2    0    0   NA
## [3,]    0   NA    1   NA    1    2    2   NA   NA   NA    2   NA    2
## [4,]   NA    0    0    2    0    2    2    0    0    0    0   NA    2
## [5,]    0    0    1    2   NA    2    2    2    0    2   NA    2    2
## [6,]    2   NA    2   NA    0    0   NA    2    0    0    2    1    1
## [7,]    2    0   NA   NA    1   NA    0    0   NA    2    0    1    1
## [8,]    2    2   NA    0   NA    2    0   NA    2   NA   NA    1    1
## [9,]   NA    0   NA   NA   NA   NA    2   NA    1    1    2    0    0
## [10,]   0    2    0   NA    2    0    1   NA    1    1    0    2    0
## [11,]    2    0    1    0    0   NA    1    0    2   NA    0    2    0
## [12,]   NA    0    0   NA    2   NA    1    1   NA    0    0    1    1
## [13,]    0    0    0   NA    0   NA    0   NA    2    1   NA    2    2
## [14,]    1    2    0    1   NA   NA    0    2   NA   NA    0    0    2
## [15,]    1    0    1    1   NA    2    1    0    2    0    2    2   NA
## [16,]    2   NA    1   NA    0    0    0    2    0    2    2    2   NA
## [17,]    1    0    0    1   NA    0   NA    0    0   NA    0    2    1
## [18,]    1    0    2    2   NA   NA   NA    0   NA    1    2    2    2
## [19,]    2    0    2   NA    1    0    2    1   NA    1    0    0    0
## [20,]    2   NA   NA    0   NA    2    1   NA    0   NA    0    1    0
## [21,]   NA    0    1    0    2    0    0    0    1    0    1    2   NA
## [22,]   NA    1    0    2    1   NA   NA   NA    1    0    0    1   NA
## [23,]    2    0    1   NA    0   NA    1    0    1    0   NA    0    0
## [24,]    0    0    0    2    1    0    0    0    2   NA   NA    2    0
## [25,]   NA    2    2   NA   NA   NA    0   NA    0    2    0    0    0
## [26,]   NA   NA    1    2    0    0    0    1    1    0   NA    0    0
## [27,]   NA    0   NA   NA   NA    2    1   NA    0   NA    1    2    1
## [28,]   NA   NA    2    0    0   NA    0    2    0    0    1   NA    0
## [29,]    0    0    2    0   NA    1   NA   NA    2    2    1   NA    2
## [30,]    0    0   NA    1    0    0    0   NA    0    0   NA   NA   NA
## [31,]    1   NA    0    0    0    0    0   NA    0   NA    2    2    2
## [32,]    0    2    2    1   NA   NA   NA    2    0   NA   NA   NA    2
## [33,]    1    0   NA   NA    1    0   NA    0    0   NA   NA    1   NA
## [34,]    1    1    0    0    0    0    2    0   NA    2   NA    0    1
## [35,]    0    0    0   NA    0    0    0   NA    2   NA    2    1    0
## [36,]    0    2   NA    0    0    1   NA   NA   NA   NA    1    0    0
## [37,]    0    2    0   NA    1    1    1    2    0    0    2    0    0
## [38,]    0   NA    0    2    2    0   NA    0   NA   NA    0    0    0
## [39,]    0    0   NA    0    0    0    0    2   NA   NA   NA   NA   NA
## [40,]    0   NA   NA    1    1   NA   NA    0    1    0    0    1    2
## [41,]   NA    0    1    0    0    0    2   NA    1    1    2    0    0
## [42,]    1    0    0    0    1   NA    0    0    0    2   NA    0    0
## [43,]    1   NA    1   NA    2    0    1    0    1    1    0    2    2
## [44,]   NA    0    0    0    0    0   NA   NA    1   NA    2    0    0
## [45,]   NA   NA   NA    1    1   NA   NA   NA    1   NA    0   NA   NA
## [46,]    1    0   NA   NA    1   NA    0    0    0    2   NA   NA   NA
## [47,]    0    2    0    0    0    2   NA   NA   NA    0   NA    0    0
## [48,]   NA    0    1    1   NA   NA    1    0   NA   NA    0    0   NA
```

## [49,]	0	0	2	NA	0	NA	2	2	0	NA	1	0	0
## [50,]	NA	0	0	0	0	2	0	1	0	NA	NA	1	0
##	[,14]	[,15]	[,16]	[,17]	[,18]	[,19]	[,20]	[,21]	[,22]	[,23]	[,24]	[,25]	
## [1,]	0	NA	2	NA	1	1	1	0	0	0	1	0	
## [2,]	1	0	NA	NA	0	2	1	2	NA	1	1	NA	
## [3,]	1	1	1	0	1	NA	2	2	NA	0	0	0	
## [4,]	2	0	0	1	1	NA	NA	2	1	1	0	2	
## [5,]	1	NA	0	2	1	1	0	NA	0	NA	0	0	
## [6,]	0	0	2	1	2	2	NA	1	NA	NA	0	1	
## [7,]	2	0	2	NA	NA	0	0	2	1	2	NA	NA	
## [8,]	1	0	1	0	1	0	0	2	NA	0	2	NA	
## [9,]	1	2	2	0	NA	NA	2	1	NA	NA	0	NA	
## [10,]	1	NA	NA	1	NA	2	0	2	NA	2	2	0	
## [11,]	2	2	NA	1	1	0	1	1	NA	NA	NA	1	
## [12,]	0	0	NA	0	1	1	2	2	NA	1	NA	2	
## [13,]	0	2	0	1	1	2	NA	0	0	0	1	NA	
## [14,]	NA	0	0	NA	NA	NA	0	NA	0	1	0	2	
## [15,]	0	2	0	0	0	1	NA	NA	NA	NA	1	0	
## [16,]	0	1	0	NA	1	NA	0	NA	0	0	NA	0	
## [17,]	2	0	2	0	1	2	0	0	1	NA	NA	2	
## [18,]	NA	2	0	0	NA	0	NA	0	NA	2	2	2	
## [19,]	NA	0	2	NA	1	NA	1	1	0	0	1	NA	
## [20,]	0	2	0	NA	NA	1	0	NA	0	NA	0	1	
## [21,]	2	1	NA	0	0	NA	2	NA	2	2	1	1	
## [22,]	1	2	2	NA	NA	0	1	1	0	0	NA	NA	
## [23,]	0	2	NA	0	0	1	0	1	0	1	0	2	
## [24,]	1	NA	2	0	NA	0	0	NA	1	NA	2	NA	
## [25,]	0	NA	2	2	0	NA	NA	2	0	2	1	2	
## [26,]	0	0	2	2	0	0	NA	1	NA	0	2	0	
## [27,]	NA	0	0	NA	NA	NA	2	2	0	1	2	0	
## [28,]	1	0	NA	0	0	0	NA	2	1	1	0	NA	
## [29,]	0	0	NA	1	0	1	0	0	NA	2	2	0	
## [30,]	2	1	NA	NA	0	NA	NA	NA	0	NA	0	2	
## [31,]	2	0	NA	2	2	0	NA	1	0	NA	1	0	
## [32,]	0	0	0	0	NA	NA	0	1	0	1	1	0	
## [33,]	2	1	0	0	1	NA	0	1	1	NA	2	NA	
## [34,]	0	0	0	2	2	0	NA	1	NA	0	0	2	
## [35,]	NA	1	1	0	2	2	0	0	1	2	0	NA	
## [36,]	2	0	0	1	0	NA	0	0	0	0	NA	0	
## [37,]	0	NA	0	1	NA	NA	2	2	NA	NA	0	NA	
## [38,]	NA	2	1	0	NA	1	0	0	0	0	0	0	
## [39,]	NA	2	2	NA	0	NA	2	NA	0	NA	0	0	
## [40,]	1	NA	NA	NA	1	1	0	0	2	NA	NA	0	
## [41,]	2	1	0	1	2	NA	NA	1	0	1	0	0	
## [42,]	2	1	2	0	NA	NA	2	0	NA	0	2	0	
## [43,]	0	0	0	1	1	1	NA	NA	0	0	1	NA	
## [44,]	0	0	NA	0	1	1	NA	1	2	NA	NA	0	
## [45,]	2	0	NA	NA	NA	0	1	1	0	1	1	NA	
## [46,]	0	1	1	NA	1	0	0	2	NA	NA	NA	NA	
## [47,]	1	NA	0	2	NA	0	1	0	NA	0	NA	0	
## [48,]	NA	1	0	NA	0	2	0	0	1	0	0	0	
## [49,]	0	0	NA	1	2	NA	0	0	0	1	0	NA	
## [50,]	NA	1	0	0	NA	NA	NA	NA	0	NA	1	0	
##	[,26]	[,27]	[,28]	[,29]	[,30]	[,31]	[,32]	[,33]	[,34]	[,35]	[,36]	[,37]	

##	[1,]	0	2	0	2	1	1	2	2	2	2	2	NA
##	[2,]	0	2	1	0	1	NA	2	NA	NA	1	0	0
##	[3,]	0	NA	0	0	2	1	0	1	2	1	0	2
##	[4,]	2	NA	2	2	0	1	NA	NA	NA	1	1	NA
##	[5,]	2	0	2	NA	NA	2	NA	0	0	1	NA	2
##	[6,]	0	NA	0	1	0	0	1	1	1	NA	2	0
##	[7,]	NA	1	0	2	0	1	1	0	NA	0	0	NA
##	[8,]	2	2	1	NA	NA	1	NA	NA	1	2	0	NA
##	[9,]	0	2	0	2	0	0	NA	NA	0	2	1	2
##	[10,]	2	0	0	NA	1	0	2	NA	2	0	0	1
##	[11,]	1	0	0	NA	NA	0	NA	2	0	2	NA	0
##	[12,]	2	NA	0	0	1	0	2	NA	2	0	NA	2
##	[13,]	2	2	NA	0	0	2	2	0	1	NA	NA	2
##	[14,]	NA	0	2	0	0	2	0	NA	0	2	0	0
##	[15,]	NA	1	1	1	1	0	NA	NA	NA	NA	NA	0
##	[16,]	NA	NA	0	1	NA	NA	2	NA	NA	2	NA	NA
##	[17,]	NA	0	2	1	0	2	0	2	0	1	0	NA
##	[18,]	0	NA	0	2	1	1	NA	0	NA	1	0	NA
##	[19,]	NA	0	NA	2	2	2	1	NA	0	NA	NA	1
##	[20,]	0	1	0	1	1	1	0	0	0	2	NA	0
##	[21,]	0	2	0	NA	0	1	0	NA	NA	NA	1	0
##	[22,]	NA	0	1	0	2	NA	0	0	2	0	0	NA
##	[23,]	0	2	1	2	NA	NA	NA	2	0	0	NA	2
##	[24,]	NA	0	2	NA	NA	0	1	2	NA	2	NA	0
##	[25,]	2	NA	0	NA	NA	0	NA	NA	0	1	1	0
##	[26,]	2	0	NA	NA	1	0	0	1	NA	0	0	0
##	[27,]	0	1	2	0	NA	NA	2	NA	NA	0	0	1
##	[28,]	NA	0	NA	0	0	0	0	1	0	NA	NA	0
##	[29,]	NA	2	1	NA	1	0	0	1	NA	NA	NA	0
##	[30,]	2	NA	0	NA	2	1	2	NA	0	1	1	1
##	[31,]	0	0	2	0	0	2	NA	0	NA	NA	NA	0
##	[32,]	0	NA	0	1	0	NA	NA	0	1	1	0	2
##	[33,]	NA	0	NA	NA	1	2	1	NA	0	NA	NA	0
##	[34,]	2	0	0	1	0	0	1	0	2	NA	NA	0
##	[35,]	NA	0	0	NA	NA	1	NA	0	0	NA	0	NA
##	[36,]	0	2	2	0	NA	NA	1	NA	2	0	NA	0
##	[37,]	0	0	0	0	0	NA	1	NA	0	0	1	0
##	[38,]	0	NA	1	0	0	2	0	2	NA	0	NA	1
##	[39,]	1	1	NA	0	0	NA	0	1	0	NA	2	1
##	[40,]	NA	0	0	NA	2	0	0	2	0	0	1	NA
##	[41,]	1	NA	0	NA	0	0	NA	0	0	0	0	NA
##	[42,]	1	0	0	NA	NA	2	0	0	1	NA	NA	NA
##	[43,]	1	NA	0	0	0	0	1	0	0	NA	NA	NA
##	[44,]	NA	2	0	NA	0	NA	0	NA	NA	NA	0	NA
##	[45,]	0	0	1	0	2	0	1	1	NA	0	0	NA
##	[46,]	1	NA	0	0	0	0	NA	1	NA	0	2	NA
##	[47,]	NA	1	0	NA	1	0	1	NA	1	0	0	0
##	[48,]	0	NA	NA	0	0	NA	0	0	2	2	2	NA
##	[49,]	0	1	2	0	1	0	0	1	0	NA	NA	NA
##	[50,]	0	1	NA	1	1	0	0	NA	NA	NA	0	0
##		[,38]	[,39]	[,40]	[,41]	[,42]	[,43]	[,44]	[,45]	[,46]	[,47]	[,48]	[,49]
##	[1,]	2	NA	0	2	2	NA	NA	NA	2	NA	2	NA
##	[2,]	2	2	1	2	1	0	NA	2	0	1	0	0
##	[3,]	NA	NA	2	NA	0	2	0	NA	0	2	NA	0

##	[4,]	1	1	0	0	2	NA	2	0	0	1	0	0
##	[5,]	NA	NA	0	2	NA	0	1	2	NA	NA	0	NA
##	[6,]	0	NA	1	NA	0	2	2	0	2	0	0	0
##	[7,]	0	NA	NA	2	2	1	0	2	0	2	2	1
##	[8,]	0	2	1	NA	2	NA	2	0	NA	0	0	0
##	[9,]	0	1	1	2	2	NA	0	0	0	2	1	NA
##	[10,]	0	0	0	2	0	NA	NA	2	0	0	NA	1
##	[11,]	NA	0	2	2	0	1	2	NA	0	1	NA	2
##	[12,]	1	0	NA	0	0	2	0	2	2	0	NA	NA
##	[13,]	0	NA	2	1	NA	NA	NA	NA	NA	1	NA	2
##	[14,]	2	0	0	2	2	1	2	1	2	0	1	NA
##	[15,]	0	NA	2	0	1	2	2	0	2	0	1	0
##	[16,]	1	NA	2	2	0	2	2	0	0	NA	0	2
##	[17,]	0	0	NA	NA	1	1	NA	NA	1	2	0	0
##	[18,]	NA	NA	0	0	0	1	0	0	1	1	0	1
##	[19,]	0	1	NA	0	1	NA	2	0	NA	0	2	NA
##	[20,]	NA	0	2	NA	NA	2	2	1	2	NA	2	0
##	[21,]	1	0	0	2	0	0	0	NA	1	0	NA	2
##	[22,]	2	0	NA	1	NA	1	0	1	NA	0	2	0
##	[23,]	NA	0	0	NA	NA	2	0	0	NA	2	NA	2
##	[24,]	NA	1	1	NA	NA	NA	0	1	0	1	0	1
##	[25,]	NA	0	0	2	NA	NA	0	1	1	NA	0	1
##	[26,]	0	1	NA	1	NA	0	NA	NA	0	2	2	2
##	[27,]	0	0	0	1	1	1	0	0	NA	1	0	1
##	[28,]	2	2	1	2	NA	1	NA	1	NA	1	1	1
##	[29,]	2	0	0	0	0	NA	2	NA	0	0	NA	0
##	[30,]	NA	0	0	0	1	NA	2	1	NA	2	0	2
##	[31,]	2	NA	0	0	1	NA	NA	NA	NA	2	0	NA
##	[32,]	NA	0	0	NA	2	1	1	1	0	1	0	0
##	[33,]	NA	NA	1	2	NA	0	1	0	NA	2	0	NA
##	[34,]	0	NA	1	0	0	NA	2	0	NA	0	0	0
##	[35,]	0	NA	2	1	NA	0	2	2	1	NA	0	NA
##	[36,]	NA	2	NA	2	NA	NA	0	1	1	2	0	1
##	[37,]	NA	1	0	NA	0	1	0	1	2	1	0	NA
##	[38,]	2	0	NA	1	NA	1	1	0	0	0	NA	2
##	[39,]	1	NA	2	NA	0	0	0	1	2	1	0	NA
##	[40,]	NA	0	0	0	0	1	0	1	NA	0	2	0
##	[41,]	0	2	NA	NA	NA	NA	2	0	1	0	0	NA
##	[42,]	0	NA	NA	NA	1	0	0	0	NA	2	NA	0
##	[43,]	NA	0	NA	NA	1	NA	0	1	NA	NA	2	NA
##	[44,]	1	NA	2	1	2	0	1	NA	0	2	0	1
##	[45,]	1	0	2	NA	NA	1	0	NA	0	0	0	1
##	[46,]	0	0	2	0	1	1	0	NA	0	NA	1	2
##	[47,]	2	1	0	1	0	0	NA	0	0	NA	0	0
##	[48,]	0	1	1	NA	NA	NA	2	NA	0	0	1	0
##	[49,]	0	NA	0	NA	0	2	NA	0	0	0	NA	NA
##	[50,]	0	0	0	NA	0	0	2	NA	1	NA	0	2
##	[,50]												
##	[1,]	NA											
##	[2,]	2											
##	[3,]	2											
##	[4,]	0											
##	[5,]	0											
##	[6,]	2											

```
## [7,] NA
## [8,] 0
## [9,] 1
## [10,] 0
## [11,] 0
## [12,] NA
## [13,] NA
## [14,] 0
## [15,] 0
## [16,] 1
## [17,] 1
## [18,] 0
## [19,] NA
## [20,] 1
## [21,] NA
## [22,] 2
## [23,] NA
## [24,] 2
## [25,] 0
## [26,] 2
## [27,] NA
## [28,] 2
## [29,] NA
## [30,] 1
## [31,] NA
## [32,] 1
## [33,] 2
## [34,] 0
## [35,] NA
## [36,] 0
## [37,] NA
## [38,] 1
## [39,] 1
## [40,] 2
## [41,] NA
## [42,] 1
## [43,] NA
## [44,] NA
## [45,] 1
## [46,] NA
## [47,] 2
## [48,] NA
## [49,] NA
## [50,] 0
```

- We will now learn the `apply` function. This is a handy function that saves writing for loops which should be eschewed in R. Use the `apply` function to compute a vector whose entries are the standard deviation of each row. Use the `apply` function to compute a vector whose entries are the standard deviation of each column. Be careful about the NA's! This should be one line (for each).

```
#TO-DO
apply(R, MARGIN=1, sd, na.rm=TRUE)
```

```
## [1] 0.81814773849 0.79042848102 0.80752760964 0.76228674466 0.85588532090
## [6] 0.83931724711 0.77385436273 0.86380197161 0.65678957743 0.87423435890
```

```
## [11] 0.79292748543 0.91079385952 0.88471806011 0.73111868704 0.75377836144
## [16] 0.87423435890 0.82139397188 0.76997217018 0.90517714369 0.78629127444
## [21] 0.81167944991 0.83190353531 0.78107876275 0.80773697723 0.81096094706
## [26] 0.82802958970 0.81649658093 0.85993941549 0.87255059327 0.86694134841
## [31] 0.91390769377 0.85512059455 0.87785814380 0.89294371875 0.76635604473
## [36] 0.79311553891 0.83029750053 0.89928422716 0.84166062230 0.87255059327
## [41] 0.65991201760 0.75134288380 0.92309308325 0.75337080350 0.84492824744
## [46] 0.89375544686 0.86380197161 0.84281592351 0.69902529542 0.83816526318
```

```
apply(R, MARGIN=2, sd, na.rm=TRUE)
```

```
## [1] 0.79681907289 0.82226112660 0.79471941424 0.84242353917 0.73192505471
## [6] 0.92979359485 0.83017419208 0.90640641031 0.78629127444 0.88473646963
## [11] 0.90097985876 0.87119134464 0.88013003579 0.85325774564 0.82814879686
## [16] 0.92023116488 0.76477052086 0.71649772083 0.79243737244 0.83816526318
## [21] 0.81069104658 0.66286796528 0.79176634141 0.80985828720 0.88359983200
## [26] 0.88593112011 0.86556706800 0.82498724166 0.81649658093 0.74293796182
## [31] 0.82553708032 0.80622577483 0.81720015416 0.85391256383 0.84611411223
## [36] 0.77385436273 0.83279552538 0.85993941549 0.75377836144 0.86388008365
## [41] 0.87904907299 0.81867681600 0.76500148544 0.94428103161 0.74907350181
## [46] 0.83212797981 0.85297368222 0.82061824559 0.84515425473 0.84002688129
```

```
apply(R, MARGIN=2, FUN = function(x){})
```

```
## NULL
```

- Use the `apply` function to compute a vector whose entries are the count of entries that are 1 or 2 in each column. This should be one line.

```
#TO-DO
```

```
?apply
```

```
apply(R, MARGIN = 2, FUN = function(x){length(x[(x==1|x==2) & !is.na(x)])})
```

```
## [1] 20 10 20 16 16 13 19 15 18 17 17 23 18 23 21 17 17 22 18 15 26 11 18 21 13
## [26] 16 18 17 15 19 19 19 16 14 18 12 13 16 13 20 22 18 22 21 19 16 23 14 20 20
```

- Use the `split` function to create a list whose keys are the column number and values are the vector of the columns. Look at the last example in the documentation `?split`.

```
#TO-DO
```

```
?split
```

```
split(R, col(R))
```

```
## $`1`
## [1] 1 NA NA 0 1 1 1 NA NA 2 0 1 0 0 NA 2 2 NA 1 NA 1 2 0 1 1
## [26] NA 0 0 2 NA 2 2 0 NA 0 NA 0 0 NA NA NA 1 0 0 0 1 2 0 1 0
##
## $`2`
## [1] 0 0 1 2 0 0 0 0 0 0 NA NA 2 2 0 0 0 0 2 NA 0 NA 0 1 0
## [26] 0 0 0 2 0 NA 0 NA 2 NA 0 0 0 NA 0 NA 0 0 2 0 0 NA 0 NA 2
##
## $`3`
## [1] 0 NA 0 0 2 0 NA 0 0 NA 0 0 0 0 1 1 2 1 0 2 0 NA 2 0 1
## [26] 1 NA 2 NA 0 1 1 1 2 NA 0 NA 0 1 NA NA NA 1 2 0 1 2 0 1 NA
##
## $`4`
## [1] 1 NA 2 NA 2 2 NA 2 0 NA 2 0 NA 0 1 NA NA 0 1 0 0 0 NA 0 1
## [26] 0 0 0 0 NA NA 0 NA NA 1 0 1 NA 2 NA 1 NA 2 1 2 0 NA NA NA 0
```

```

##
## $`5`
## [1] NA NA 1 1 NA 0 1 0 0 1 2 0 2 0 NA 0 1 0 NA 0 1 NA 0 0 NA
## [26] 2 0 NA NA 2 0 0 1 NA 1 0 0 0 0 NA 1 1 NA NA 1 NA 0 0 2 0
##
## $`6`
## [1] 0 2 NA 1 NA NA 0 2 2 NA 0 0 0 2 NA NA 0 0 NA NA NA 2 NA 0 2
## [26] 0 0 1 2 NA 0 NA 2 NA NA 0 0 NA 0 NA NA NA 2 NA 0 2 0 0 0 1
##
## $`7`
## [1] NA 1 NA 1 NA 2 NA 2 0 0 NA 0 1 NA 1 1 2 2 0 0 0 1 2 2 1
## [26] 0 0 NA 0 1 0 1 2 0 NA NA 0 0 0 2 NA 0 2 NA 0 0 NA 0 1 NA
##
## $`8`
## [1] 0 NA NA 2 0 1 0 0 1 0 0 NA NA NA 0 0 1 NA 2 2 0 NA 2 0 0
## [26] 0 2 NA NA 1 2 0 NA NA 0 NA NA NA 1 NA NA 0 2 2 0 2 2 NA 0 NA
##
## $`9`
## [1] 0 0 1 0 NA NA 0 0 0 NA NA 0 1 NA NA 1 NA 1 NA 0 0 0 0 NA 2
## [26] 1 NA 2 2 NA 0 2 NA 0 1 1 0 2 1 1 1 0 0 0 2 NA 0 2 1 NA
##
## $`10`
## [1] NA NA 0 0 1 2 NA 0 NA 2 NA NA 1 0 NA 0 1 1 NA 0 2 NA NA 2 0
## [26] 0 NA 2 NA 0 2 NA NA 2 0 NA 0 1 0 1 NA 2 2 NA NA 2 0 NA 1 NA
##
## $`11`
## [1] 0 1 0 2 2 0 NA 0 NA 0 0 2 0 NA 0 NA 0 2 0 1 NA 0 1 NA 2
## [26] 1 NA 1 NA 0 2 0 2 0 0 2 NA NA NA 2 0 NA NA NA NA 0 2 2 0 1
##
## $`12`
## [1] 2 2 1 0 2 0 1 NA 1 1 0 2 2 0 0 0 0 0 0 NA 0 1 0 0 2
## [26] 2 NA NA 1 1 2 2 NA 0 1 0 NA 2 0 0 NA NA 2 NA 2 NA 1 1 2 0
##
## $`13`
## [1] 1 1 NA 0 2 NA NA 2 0 1 0 2 0 0 NA 0 0 0 2 0 0 0 0 1 NA
## [26] NA NA 2 1 1 NA 0 2 0 2 0 NA 2 0 0 NA NA 2 2 0 0 1 0 2 0
##
## $`14`
## [1] 2 NA 1 0 NA 1 2 2 NA 2 NA 2 1 1 NA 0 NA 2 NA 1 2 0 0 0 0
## [26] 2 NA 0 1 0 0 2 1 0 1 0 2 0 0 1 2 0 1 0 1 0 0 NA 0 2
##
## $`15`
## [1] 0 0 2 NA 2 0 1 0 1 0 2 0 NA NA 1 2 0 1 0 0 1 2 0 0 2
## [26] 1 2 0 0 0 1 2 1 NA NA 0 1 2 0 2 0 1 NA 0 NA NA 0 1 0 0
##
## $`16`
## [1] 2 0 2 0 0 NA 0 0 0 2 1 NA NA 0 0 NA 2 0 0 NA 2 0 NA 0 0
## [26] NA 2 NA 1 NA 0 NA 1 2 NA NA NA 0 2 2 NA 1 0 0 2 2 2 1 0 0
##
## $`17`
## [1] 0 NA NA 1 0 NA 0 1 0 NA 0 2 1 2 NA 0 NA 1 NA 0 0 NA 1 2 0
## [26] 0 NA 1 0 0 NA 1 0 2 NA 0 NA 1 2 0 NA NA 2 0 0 NA 1 0 1 1
##
## $`18`

```

```

## [1] 1 NA NA NA NA 0 1 1 NA NA NA 2 NA NA 0 0 1 2 NA 0 NA NA 2 2 0
## [26] 0 0 0 1 1 1 1 1 0 1 1 0 1 0 NA NA 1 1 NA NA 1 2 2 1 0
##
## $`19`
## [1] 2 NA 0 NA 0 2 NA NA NA 0 1 0 2 0 2 1 NA NA NA 0 NA 1 NA 0 1
## [26] NA NA 1 0 1 NA 0 NA NA 1 1 NA 2 0 NA 0 0 1 NA 0 1 2 2 1 NA
##
## $`20`
## [1] 0 2 1 2 NA 1 0 NA NA 0 0 NA 0 1 0 0 1 NA 0 NA 2 0 0 NA NA
## [26] 2 2 0 0 2 0 1 2 NA 0 NA NA NA NA 2 1 0 0 0 0 1 NA 0 NA 0
##
## $`21`
## [1] 0 2 1 2 0 2 1 2 NA 2 0 1 2 0 0 1 1 1 NA 2 0 NA 0 1 NA
## [26] NA NA 0 2 2 NA 1 2 2 0 1 NA 0 1 1 1 2 NA 1 NA 0 1 0 NA 0
##
## $`22`
## [1] 1 0 0 NA NA NA 1 1 0 1 0 0 NA NA 1 0 0 0 0 1 NA 0 0 NA NA
## [26] 2 0 NA NA NA 0 NA NA 0 2 2 0 0 NA NA 0 NA 0 0 1 0 NA 1 0 0
##
## $`23`
## [1] NA 1 0 NA 2 1 NA 1 NA 2 0 NA 2 0 0 1 0 1 1 1 0 NA 1 0 NA
## [26] 2 NA 2 0 1 0 NA 0 2 NA NA NA 0 0 NA 1 NA NA 1 NA 0 NA 2 0 0
##
## $`24`
## [1] NA 2 NA 0 2 1 2 0 1 NA 0 1 2 NA 0 0 1 0 0 0 2 0 0 0 1
## [26] 1 0 2 2 NA NA NA 0 1 NA NA 0 1 2 0 1 NA 0 1 2 1 0 0 1 NA
##
## $`25`
## [1] 2 0 NA NA 2 NA NA 2 0 NA 0 0 0 0 0 2 NA 0 2 NA 0 1 NA 2 0
## [26] 1 0 0 NA 2 0 1 0 2 0 0 2 NA 0 NA NA NA 0 0 NA 0 1 NA NA 0
##
## $`26`
## [1] NA 0 NA 0 0 0 NA 2 0 NA 0 0 2 NA 0 0 NA 1 NA NA 1 0 0 2 NA
## [26] 0 1 NA 2 2 NA 1 0 2 NA NA 2 2 2 0 0 1 2 0 NA 0 0 NA 1 0
##
## $`27`
## [1] 0 1 0 0 NA 2 0 NA 1 1 NA 0 0 1 NA 2 0 NA 0 0 0 1 1 0 1
## [26] 2 1 2 2 NA NA 0 NA NA 0 2 NA 2 0 2 0 NA 0 NA 0 2 NA 0 NA 2
##
## $`28`
## [1] 2 2 1 0 0 1 NA 2 NA 0 1 2 0 0 NA 1 NA 0 2 NA 0 0 2 0 1
## [26] 0 NA 1 1 0 0 0 0 0 0 0 0 0 NA NA 0 1 0 2 0 2 0 0 0 2
##
## $`29`
## [1] 1 0 0 0 2 0 NA 2 1 2 0 0 NA NA 0 2 2 NA 0 0 NA 1 0 1 1
## [26] NA 0 NA NA 0 1 NA 0 NA NA NA NA 0 NA 2 0 0 NA 1 NA 2 1 NA 0 0
##
## $`30`
## [1] 0 NA 2 0 1 1 1 0 1 0 0 0 1 1 0 NA 2 0 0 0 NA 1 1 0 1
## [26] 0 0 1 NA 1 NA NA 2 NA 2 0 2 0 1 0 2 0 NA 0 NA 1 0 NA 0 NA
##
## $`31`
## [1] 2 NA NA NA 1 NA 2 1 0 1 2 2 0 0 NA NA 2 0 2 0 2 1 0 0 0
## [26] 1 NA 0 1 0 NA 0 1 0 0 NA 1 2 0 0 0 0 2 NA 0 1 0 1 0 NA

```

```

##
## $`32`
## [1] 0 2 0 1 NA 2 1 NA 0 1 0 NA 2 1 0 NA 1 NA 0 0 0 0 1 NA
## [26] 0 0 0 NA 2 2 NA 0 NA 0 0 2 2 0 NA 1 NA NA NA 1 2 1 NA 1 1
##
## $`33`
## [1] 2 NA 0 NA 0 NA NA NA NA 0 2 0 NA NA 0 2 NA 0 NA 1 0 0 1 0 NA
## [26] NA 1 1 NA NA NA 2 1 NA 2 NA NA 0 1 NA 1 1 0 0 2 2 1 0 0 NA
##
## $`34`
## [1] 0 NA 2 0 NA NA 0 NA NA NA NA NA 2 1 2 0 0 0 0 0 1 0 0 2 NA
## [26] NA 0 NA 1 2 NA 0 2 0 0 NA 0 1 NA 0 NA NA 0 1 NA 2 1 0 0 2
##
## $`35`
## [1] 1 0 0 0 1 1 NA 1 NA 0 0 NA 0 0 2 0 NA 0 2 NA NA 2 NA NA NA
## [26] NA NA NA 2 0 2 2 1 1 0 NA 1 NA 0 2 0 0 1 1 2 2 NA NA NA 0
##
## $`36`
## [1] 0 0 0 1 0 0 NA 1 0 0 NA NA 0 0 2 NA NA 0 0 NA NA NA NA NA NA
## [26] 1 2 NA 0 NA NA NA 0 1 1 0 1 NA 0 1 0 2 NA 0 NA 2 2 0 NA NA
##
## $`37`
## [1] NA 1 NA 0 NA 0 0 NA 0 NA 1 0 1 0 NA 2 1 NA 0 0 NA 0 NA 0 0
## [26] 0 1 0 NA 2 NA 0 2 0 NA NA 1 2 0 2 NA NA 2 2 0 NA 0 NA NA 0
##
## $`38`
## [1] 0 0 2 NA NA 2 NA 1 0 0 2 2 0 2 0 NA 0 0 2 2 0 NA 0 0 0
## [26] 1 1 2 0 1 1 NA NA NA NA 1 NA 0 0 0 1 0 NA NA NA 2 0 0 NA NA
##
## $`39`
## [1] 0 0 0 1 NA 2 NA 1 0 NA 0 NA 0 1 1 0 1 2 0 2 NA 0 NA NA NA
## [26] 0 NA 0 2 0 NA 0 NA 0 0 NA 0 NA 1 1 0 0 NA 0 1 NA NA NA 0 2
##
## $`40`
## [1] NA 0 NA 0 0 1 1 0 0 NA NA 0 0 0 1 0 NA NA 0 1 NA 2 0 1 2
## [26] 0 2 0 1 NA 2 2 2 0 0 2 0 2 NA 1 2 2 0 0 1 0 1 2 NA NA
##
## $`41`
## [1] NA 1 1 NA 0 2 2 0 NA 2 1 0 2 1 NA NA 0 NA 2 2 NA NA NA 0 0
## [26] 2 NA 0 NA 0 2 2 NA 2 0 1 0 1 1 2 NA 0 2 NA NA 2 NA 1 NA 2
##
## $`42`
## [1] 1 1 NA 0 0 1 NA 2 0 2 NA 1 0 0 NA NA 1 NA 2 NA 1 NA 0 0 1
## [26] 0 0 0 2 0 0 0 0 NA 0 2 1 NA NA 2 NA 1 NA 2 NA 2 0 NA 1 NA
##
## $`43`
## [1] 1 1 1 1 1 0 0 NA 0 1 1 NA NA 0 NA 2 NA NA 1 1 0 2 2 NA 2
## [26] 0 0 NA NA 2 2 1 2 NA 1 0 NA NA 0 NA 1 1 0 1 NA NA 2 0 NA NA
##
## $`44`
## [1] NA 0 0 0 0 NA 1 2 2 0 1 NA NA NA 2 0 2 2 2 NA 0 2 NA 2 2
## [26] 0 0 2 2 0 2 2 0 0 0 1 2 NA NA 0 0 0 1 1 0 NA 2 2 0 0
##
## $`45`

```

```
## [1] NA 0 1 1 0 2 0 0 NA 2 0 NA 2 0 NA 0 0 0 1 1 0 1 0 0 0
## [26] NA 1 NA 0 2 0 NA NA 1 1 NA 1 NA NA 0 NA NA 2 1 1 NA 0 2 1 1
##
## $`46`
## [1] 1 NA NA 2 1 0 NA 0 1 0 0 NA 0 0 0 NA NA 1 2 NA NA 2 0 NA 2
## [26] 1 2 0 NA 2 0 0 0 1 NA 0 NA NA 0 0 0 0 NA 0 0 2 2 1 NA 1
##
## $`47`
## [1] 2 1 0 1 1 1 2 1 NA 2 0 2 0 NA 0 2 0 0 0 1 2 NA 0 0 0
## [26] 0 1 0 0 0 NA 1 2 NA 0 2 2 1 2 2 0 NA NA 1 1 NA 0 NA NA 2
##
## $`48`
## [1] 0 0 2 0 0 0 0 0 0 0 2 NA 0 NA 0 1 NA 2 0 1 1 NA 2 NA 0 1
## [26] NA 0 NA 0 NA 0 NA NA 0 2 0 0 NA 2 1 0 1 0 0 0 2 0 0 2 0
##
## $`49`
## [1] 0 1 0 NA 1 0 NA 0 2 1 2 NA 1 0 0 2 NA NA NA 1 0 0 NA 0 0
## [26] 2 NA 0 0 NA 2 2 0 1 0 1 2 2 2 NA 1 2 NA 0 1 NA 0 NA NA 1
##
## $`50`
## [1] 1 NA 2 NA 0 2 2 0 0 NA 1 NA 0 2 NA NA NA NA 0 2 1 1 NA 0 0
## [26] NA 1 NA 0 NA 1 0 2 0 2 NA 1 NA 2 1 1 NA 0 1 2 NA 2 NA NA 0
```

- In one statement, use the `lapply` function to create a list whose keys are the column number and values are themselves a list with keys: “min” whose value is the minimum of the column, “max” whose value is the maximum of the column, “pct_missing” is the proportion of missingness in the column and “first_NA” whose value is the row number of the first time the NA appears.

```
#TO-DO
?lapply
lapply(split(R, col(R)), FUN = function(x){as.list(c(min(x, na.rm = TRUE), max(x, na.rm = TRUE), pct_mi

## $`1`
## $`1`[[1]]
## [1] 0
##
## $`1`[[2]]
## [1] 2
##
## $`1`$pct_missing
## [1] 0.28
##
##
## $`2`
## $`2`[[1]]
## [1] 0
##
## $`2`[[2]]
## [1] 2
##
## $`2`$pct_missing
## [1] 0.22
##
##
## $`3`
```



```

## $`3`[[1]]
## [1] 0
##
## $`3`[[2]]
## [1] 2
##
## $`3`$pct_missing
## [1] 0.24
##
##
## $`4`
## $`4`[[1]]
## [1] 0
##
## $`4`[[2]]
## [1] 2
##
## $`4`$pct_missing
## [1] 0.36
##
##
## $`5`
## $`5`[[1]]
## [1] 0
##
## $`5`[[2]]
## [1] 2
##
## $`5`$pct_missing
## [1] 0.28
##
##
## $`6`
## $`6`[[1]]
## [1] 0
##
## $`6`[[2]]
## [1] 2
##
## $`6`$pct_missing
## [1] 0.38
##
##
## $`7`
## $`7`[[1]]
## [1] 0
##
## $`7`[[2]]
## [1] 2
##
## $`7`$pct_missing
## [1] 0.26
##
##

```

```

## $`8`
## $`8`[[1]]
## [1] 0
##
## $`8`[[2]]
## [1] 2
##
## $`8`$pct_missing
## [1] 0.36
##
##
## $`9`
## $`9`[[1]]
## [1] 0
##
## $`9`[[2]]
## [1] 2
##
## $`9`$pct_missing
## [1] 0.28
##
##
## $`10`
## $`10`[[1]]
## [1] 0
##
## $`10`[[2]]
## [1] 2
##
## $`10`$pct_missing
## [1] 0.4
##
##
## $`11`
## $`11`[[1]]
## [1] 0
##
## $`11`[[2]]
## [1] 2
##
## $`11`$pct_missing
## [1] 0.3
##
##
## $`12`
## $`12`[[1]]
## [1] 0
##
## $`12`[[2]]
## [1] 2
##
## $`12`$pct_missing
## [1] 0.2
##
##

```

```

##
## $`13`
## $`13`[[1]]
## [1] 0
##
## $`13`[[2]]
## [1] 2
##
## $`13`$pct_missing
## [1] 0.22
##
##
## $`14`
## $`14`[[1]]
## [1] 0
##
## $`14`[[2]]
## [1] 2
##
## $`14`$pct_missing
## [1] 0.18
##
##
## $`15`
## $`15`[[1]]
## [1] 0
##
## $`15`[[2]]
## [1] 2
##
## $`15`$pct_missing
## [1] 0.16
##
##
## $`16`
## $`16`[[1]]
## [1] 0
##
## $`16`[[2]]
## [1] 2
##
## $`16`$pct_missing
## [1] 0.28
##
##
## $`17`
## $`17`[[1]]
## [1] 0
##
## $`17`[[2]]
## [1] 2
##
## $`17`$pct_missing
## [1] 0.3

```

```

##
##
## $`18`
## $`18`[[1]]
## [1] 0
##
## $`18`[[2]]
## [1] 2
##
## $`18`$pct_missing
## [1] 0.32
##
##
## $`19`
## $`19`[[1]]
## [1] 0
##
## $`19`[[2]]
## [1] 2
##
## $`19`$pct_missing
## [1] 0.38
##
##
## $`20`
## $`20`[[1]]
## [1] 0
##
## $`20`[[2]]
## [1] 2
##
## $`20`$pct_missing
## [1] 0.3
##
##
## $`21`
## $`21`[[1]]
## [1] 0
##
## $`21`[[2]]
## [1] 2
##
## $`21`$pct_missing
## [1] 0.22
##
##
## $`22`
## $`22`[[1]]
## [1] 0
##
## $`22`[[2]]
## [1] 2
##
## $`22`$pct_missing

```

```

## [1] 0.34
##
##
## $`23`
## $`23`[[1]]
## [1] 0
##
## $`23`[[2]]
## [1] 2
##
## $`23`$pct_missing
## [1] 0.34
##
##
## $`24`
## $`24`[[1]]
## [1] 0
##
## $`24`[[2]]
## [1] 2
##
## $`24`$pct_missing
## [1] 0.22
##
##
## $`25`
## $`25`[[1]]
## [1] 0
##
## $`25`[[2]]
## [1] 2
##
## $`25`$pct_missing
## [1] 0.32
##
##
## $`26`
## $`26`[[1]]
## [1] 0
##
## $`26`[[2]]
## [1] 2
##
## $`26`$pct_missing
## [1] 0.3
##
##
## $`27`
## $`27`[[1]]
## [1] 0
##
## $`27`[[2]]
## [1] 2
##

```

```

## $`27`$pct_missing
## [1] 0.28
##
##
## $`28`
## $`28`[[1]]
## [1] 0
##
## $`28`[[2]]
## [1] 2
##
## $`28`$pct_missing
## [1] 0.16
##
##
## $`29`
## $`29`[[1]]
## [1] 0
##
## $`29`[[2]]
## [1] 2
##
## $`29`$pct_missing
## [1] 0.34
##
##
## $`30`
## $`30`[[1]]
## [1] 0
##
## $`30`[[2]]
## [1] 2
##
## $`30`$pct_missing
## [1] 0.22
##
##
## $`31`
## $`31`[[1]]
## [1] 0
##
## $`31`[[2]]
## [1] 2
##
## $`31`$pct_missing
## [1] 0.22
##
##
## $`32`
## $`32`[[1]]
## [1] 0
##
## $`32`[[2]]
## [1] 2

```

```

##
## $\`32`$pct_missing
## [1] 0.28
##
##
## $\`33`
## $\`33`[[1]]
## [1] 0
##
## $\`33`[[2]]
## [1] 2
##
## $\`33`$pct_missing
## [1] 0.4
##
##
## $\`34`
## $\`34`[[1]]
## [1] 0
##
## $\`34`[[2]]
## [1] 2
##
## $\`34`$pct_missing
## [1] 0.34
##
##
## $\`35`
## $\`35`[[1]]
## [1] 0
##
## $\`35`[[2]]
## [1] 2
##
## $\`35`$pct_missing
## [1] 0.34
##
##
## $\`36`
## $\`36`[[1]]
## [1] 0
##
## $\`36`[[2]]
## [1] 2
##
## $\`36`$pct_missing
## [1] 0.4
##
##
## $\`37`
## $\`37`[[1]]
## [1] 0
##
##
## $\`37`[[2]]

```

```

## [1] 2
##
## $`37`$pct_missing
## [1] 0.36
##
##
## $`38`
## $`38`[[1]]
## [1] 0
##
## $`38`[[2]]
## [1] 2
##
## $`38`$pct_missing
## [1] 0.3
##
##
## $`39`
## $`39`[[1]]
## [1] 0
##
## $`39`[[2]]
## [1] 2
##
## $`39`$pct_missing
## [1] 0.34
##
##
## $`40`
## $`40`[[1]]
## [1] 0
##
## $`40`[[2]]
## [1] 2
##
## $`40`$pct_missing
## [1] 0.22
##
##
## $`41`
## $`41`[[1]]
## [1] 0
##
## $`41`[[2]]
## [1] 2
##
## $`41`$pct_missing
## [1] 0.34
##
##
## $`42`
## $`42`[[1]]
## [1] 0
##
##

```



```

## $`42`[[2]]
## [1] 2
##
## $`42`$pct_missing
## [1] 0.32
##
##
## $`43`
## $`43`[[1]]
## [1] 0
##
## $`43`[[2]]
## [1] 2
##
## $`43`$pct_missing
## [1] 0.34
##
##
## $`44`
## $`44`[[1]]
## [1] 0
##
## $`44`[[2]]
## [1] 2
##
## $`44`$pct_missing
## [1] 0.2
##
##
## $`45`
## $`45`[[1]]
## [1] 0
##
## $`45`[[2]]
## [1] 2
##
## $`45`$pct_missing
## [1] 0.28
##
##
## $`46`
## $`46`[[1]]
## [1] 0
##
## $`46`[[2]]
## [1] 2
##
## $`46`$pct_missing
## [1] 0.3
##
##
## $`47`
## $`47`[[1]]
## [1] 0

```

```
##
## $`47`[[2]]
## [1] 2
##
## $`47`$pct_missing
## [1] 0.2
##
##
## $`48`
## $`48`[[1]]
## [1] 0
##
## $`48`[[2]]
## [1] 2
##
## $`48`$pct_missing
## [1] 0.22
##
##
## $`49`
## $`49`[[1]]
## [1] 0
##
## $`49`[[2]]
## [1] 2
##
## $`49`$pct_missing
## [1] 0.28
##
##
## $`50`
## $`50`[[1]]
## [1] 0
##
## $`50`[[2]]
## [1] 2
##
## $`50`$pct_missing
## [1] 0.36
```

- Set a seed and then create a vector `v` consisting of a sample of 1,000 iid normal realizations with mean -10 and variance 100.

```
set.seed(2024)
v = rnorm(1000, mean = -10, sd = 10) # variance = 100 implies sd = 10
v
```

```
##      [1] -0.1803058864956 -5.3128495996391 -11.0797133221644 -12.1287816897121
##      [5]  1.5809845503933  2.9235484543788 -4.6535285386769 -11.2703471920428
##      [9] -22.2487541446907 -21.2135130216029 -26.7192502409290 -5.3108207021616
##     [13] -1.6781009352237 -7.0697839488779 -42.7428582255762 -11.4811075118461
##     [17] -19.2993873527463 -5.4578564843187 -26.0610724067137 -30.1147132874967
##     [21]  1.3997663396662 -6.8429704817816  0.4268610000034 -12.1298249662589
##     [25] -6.0289961835808 -25.4365250071121 -11.1819970164530 -3.1530211525039
##     [29] -30.5572231781660 -11.0942400576655 -24.3979364518451 -13.2533497272815
```

```

## [33] -15.0528088789271 2.0424903908767 -14.9897889850454 -6.1040451829392
## [37] -6.6266812513626 -14.8926155005023 -15.7077106069217 -5.7163593582371
## [41] -18.7246183720614 -10.0469101335874 -5.0462485101978 9.7281867936945
## [45] -21.9686212395264 -8.6956151888316 -6.7094035981401 -3.3184169838100
## [49] -3.6100704262892 9.4045129634858 -15.5453906473243 -17.6571105903680
## [53] -16.4669436131039 -20.2753085374284 -16.8089807146081 -24.3361066132337
## [57] -1.0450648750490 -12.3906127048479 0.0040381287092 -10.3258997983615
## [61] 5.9294707621239 -1.3980256772202 -4.6163023243592 -8.2475426218326
## [65] 6.4589814017451 -11.2888793985204 -29.9550397970437 -2.8536061059492
## [69] -5.0830043704118 -31.1881259117686 1.3131845477581 -22.0821528021287
## [73] -7.0413408140621 -9.2473130993512 -8.0383975169200 -13.1275889974135
## [77] 4.2129014680825 -3.1652944949978 -20.3885629111138 -9.8431731059745
## [81] -22.4320967834395 -0.0814101840567 -1.3603486040158 -27.5503854875554
## [85] -8.5119178148385 1.1007456035807 -11.8275291438331 -12.5518776801131
## [89] -12.5093668702369 -1.8337904073086 -22.2741533746968 -15.2123011220730
## [93] -17.0471244856794 -3.1256252981465 -4.5240710446482 -30.6585280355845
## [97] -21.6581874873459 -21.7925052730277 -10.2975935989188 1.5152179090866
## [101] 12.0866170263516 -5.5986721731277 0.4316598755119 -8.7432325507772
## [105] -22.6524945482264 -6.2563996298477 -11.9960941530021 1.5478290427735
## [109] -15.6170570982065 -21.3775660694886 -22.9337009302956 1.2481194347300
## [113] -17.0100072717359 -10.2887580967693 -16.2497488748691 -28.3036801979779
## [117] 9.0479316745350 -4.7008088504069 -12.6479094896724 -0.9285275439649
## [121] -12.6526857798807 -13.2856747101680 -23.0367636155847 -11.9728943649195
## [125] -17.1144005562322 -9.7115862232025 -20.7349485921401 -20.9179206421077
## [129] -15.6900962765166 -4.8044187442224 -19.7017495311668 -8.6648142550345
## [133] -16.4971647805493 -9.6326934922244 6.9080705752006 -18.4300123249546
## [137] -22.0077212142888 -4.7192219074841 -6.7385602710682 -12.9990907977219
## [141] -4.4826200341336 -22.4650545927955 -17.9645300009628 7.4334964989931
## [145] -1.3594867520235 -2.6879611342693 7.3516661328297 -8.6162281515496
## [149] -1.2794240206990 0.5024875250688 -21.3743873910540 -10.4365713191630
## [153] 13.9353017391610 2.4444013015039 -17.6721305739131 -39.5151475392099
## [157] -6.7076080546818 -13.0576161062312 -17.1253431140826 -7.0775219827311
## [161] 7.3894209134024 -10.0257142424582 -10.0446431663702 5.6199313824202
## [165] -13.8050595491185 -16.3288578422368 -8.6298086204745 -27.2531305864046
## [169] 2.2290580850524 11.9604004019626 -13.6852332336776 -0.7408931368001
## [173] -14.8947777660375 -7.9244764640806 -6.0187220172517 1.5603750967308
## [177] 2.4213119154568 -8.3409001745624 -2.8478230554062 -13.7840431968211
## [181] -3.0451797006465 -7.4539642204246 -3.2995869829660 -16.6485489052397
## [185] -13.7525001051724 -2.4775516591910 -0.0344484958014 -13.6030366815733
## [189] -16.7006000567486 -5.2150723887122 -15.8868120925210 -20.4524717660040
## [193] -4.4465860119758 0.6552184630196 2.8764954631915 -4.1652489860616
## [197] 2.6413132433733 -5.1158488814728 12.9654989495162 -28.1315831747814
## [201] -9.9781053874791 -5.2463566539417 -17.7774249631156 -7.9462232497621
## [205] 0.2254474171184 -20.1007681040258 -17.5193350938369 -8.5580756768530
## [209] -1.1944177122532 -5.6863872549372 -16.9094642002030 -4.4065923962857
## [213] -15.6379052057320 -37.4231751288163 -3.5731735359120 -6.0077321752495
## [217] -3.0128700567797 -4.7754292606120 -15.9652293441173 0.7961910629198
## [221] -16.5206818570561 4.9547796546571 -8.6560181489454 -10.7351641084695
## [225] -9.7372547077482 -7.6067904582373 -10.8463095909997 -5.4097891355933
## [229] -12.5589099734331 -3.8404797721617 3.4711843045988 -3.2417131672712
## [233] -12.1971482053401 -13.7285224424273 -10.3198993447715 -9.1271964043872
## [237] -10.6355535193751 -15.1565066573624 -9.2192272750745 -19.1668919768462
## [241] -7.8254123237141 0.9168949981880 3.6498500105008 1.3418420694798
## [245] -19.1022335338185 8.9166103356048 -15.6468439458776 -7.4216545160238

```

```

## [249] -21.2195524725869 -8.4688269063652 7.2913769263462 -8.2760975885274
## [253] 0.2596319970137 -12.7771822608122 -7.2793918414961 -25.7414945335862
## [257] -8.5208542548466 -11.6110041121826 -4.4759120759735 -21.5087326123939
## [261] -12.3872725222163 -13.4353666341267 -4.7053522090751 -17.5617508326285
## [265] -35.4076757969432 -3.9159166909376 -14.5473808433741 -2.7123371069030
## [269] -4.6011998516130 -0.7499225551325 -1.0514949152143 -11.7962877486120
## [273] -23.4208555950708 -18.1489249323797 -4.2845610091924 -9.6956165240828
## [277] -16.2034083120522 -25.8822838421770 -19.3543929083080 -11.1871176211003
## [281] -3.3502072879066 -7.6023117813668 3.6380597291200 -7.7646325765409
## [285] -3.8959488441810 -13.9180554538861 -17.7900079357948 -9.3606193905650
## [289] 0.9715348501290 15.3319326201348 -16.8987214590802 -1.9632114579366
## [293] 6.5778773662102 -14.1704326538974 -6.6195272487084 -2.9487321357761
## [297] 3.1251820689780 -22.2903959055421 -22.7419637679424 -8.6346785021009
## [301] -2.8879715551943 -16.3237634881880 -13.1791026260279 0.6750166236092
## [305] 5.9225516483838 -5.5064090079456 -11.0815443721089 -2.8452574892289
## [309] -8.2016533005877 3.1858368598170 -6.1035526994700 -23.3516696872942
## [313] -26.0958358576562 -18.6980742301958 19.3053462033689 -14.2158021018915
## [317] -10.3982972108911 -9.6187331173262 0.4256051189586 -16.4132760478704
## [321] -20.6846563632480 -8.0014864183576 -6.1183791805772 -16.7945065339838
## [325] 16.5717077824476 -11.6325052715059 -20.2436398624072 -3.1407578436457
## [329] -29.5281449618610 -13.6462601978833 -9.1629835138199 -16.2497141268932
## [333] 3.3842452662869 -14.7823765993881 1.3123890603547 -25.1502282813937
## [337] 3.7030178185034 -30.7515452637654 -9.9134529593649 -5.5705288143482
## [341] -16.5347403859934 -25.9085900902484 -6.7256104956540 -9.6478189267555
## [345] -2.7544479810451 5.1307498906351 -21.5059395530594 -5.2111101797383
## [349] -21.4549857516378 2.9657907014957 -14.0894173914832 -2.0350661938241
## [353] -2.7707677343796 -17.5797109364871 -30.3990965838588 -15.0888637681959
## [357] -2.7086263530102 -1.8473404183868 1.7467553659151 -21.0257720675612
## [361] -12.4534847654154 -4.5002112495953 -17.4933675212278 -16.5211396361406
## [365] -5.5818286733258 -15.2912234904135 -8.2866222842248 -8.8348494286810
## [369] -4.0946318561786 -17.1085293777947 7.4410233789987 -14.2472552750968
## [373] -14.1248522713613 -11.4924505532460 -9.5095166492796 -11.2992394392485
## [377] -12.0886205696387 -14.6289656404120 13.2543063320106 -18.4242859148177
## [381] -8.2607578794608 -12.4468438603021 -7.3554326636067 -9.6904151063418
## [385] 5.2773124922528 -6.0956433738833 -26.1494453802577 -1.7893261110343
## [389] -11.3236998467861 2.0613241061493 -9.8566235047196 -2.5653422641053
## [393] -1.0628649783560 -21.4385200066133 4.1837283138420 -4.4806434147355
## [397] -14.1430542127147 -11.5977897841701 -23.5113512458166 -11.8307848687155
## [401] 10.0043449208795 -2.1159572481764 -8.1009628684437 -18.2942633591117
## [405] -0.4204483155755 -30.6018677286537 -7.5378979197954 -11.0477211883229
## [409] -6.7766389318553 -2.9507513424649 3.9790201729547 -9.2335920366812
## [413] -1.7976033506009 -2.6743254564307 -2.6047341558846 -9.2345669076303
## [417] -20.8016351939664 -18.0443228024388 -25.5960874243616 -3.8948017822234
## [421] -27.1737292102605 5.7518077731523 -11.0187842798778 -14.3257810833947
## [425] -1.4716660278786 -12.9277100651366 4.5960465799314 -23.8444524076257
## [429] -1.9456894376918 4.1562667785803 10.9868406775059 4.4487478162307
## [433] -0.1920302400843 -19.1622711812879 -0.0831777644270 -0.1183574393984
## [437] -26.2960876100431 -18.9365567358040 -15.9125355401261 -10.1314811861530
## [441] -16.1571490030858 5.6209432739697 -12.1498625174759 -14.4502675178677
## [445] -4.4011676789614 -14.0032104090842 -22.2499377367224 -15.2308058265306
## [449] -8.1139299739298 -14.0942052797231 -21.2733475290809 -1.0751571390927
## [453] -18.5654524996882 -32.6542481431424 -2.3362794276395 -0.3712268126441
## [457] -5.3070241761462 -2.6271483643062 -22.0952630532017 2.3107795114116
## [461] -3.4631546432828 -3.6458957774769 -4.6228270095043 1.8019131907331

```

```

## [465] -25.7128554150594 -10.4319328552306 -8.6302343129058 2.9732483341246
## [469] 8.5076619146239 -15.8620522291904 -2.2635333048804 -12.0837709790459
## [473] -13.1528611984168 -10.4097460609327 -12.1549167222693 -10.3710715647018
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## [997] -29.9072101943124 -14.4756992723880 -24.2059698391557 -2.2930917492870
```

- Repeat this exercise by resetting the seed to ensure you obtain the same results.

```
set.seed(2024)
v = rnorm(1000, mean = -10, sd = 10) # variance = 100 implies sd = 10
v
```

```
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## [85] -8.5119178148385 1.1007456035807 -11.8275291438331 -12.5518776801131
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## [89] -12.5093668702369 -1.8337904073086 -22.2741533746968 -15.2123011220730
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```



```

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```

```
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## [993] -13.2968102728749 -21.9500424055907 -4.2228036029409 0.5050933898074
## [997] -29.9072101943124 -14.4756992723880 -24.2059698391557 -2.2930917492870
```

- Find the average of `v` and the standard error of `v`.

```
mean(v)
```

```
## [1] -9.9545730151
```

```
sd(v)
```

```
## [1] 9.8271506531
```

- Find the 5%ile of `v` and use the `qnorm` function to compute what it theoretically should be. Is the estimate about what is expected by theory?

The estimated 5%-ile of 'v' is quite close to the theoretical 5%-ile of the normal distribution whose values of 'v' are samples of.

```
# Estimate 5%ile using v
quantile(v, probs = 0.05)
```

```
##          5%
## -26.159964106
```

```
# Theoretical 5%ile
qnorm(0.05, mean = -10, sd = 10)
```

```
## [1] -26.44853627
```

- What is the percentile of `v` that corresponds to the value 0? What should it be theoretically? Is the estimate about what is expected by theory?

I manually computed the percentile of 'v' that corresponds to the value 0 and its close to the theoretical value.

```
# Percentile of 'v' that corresponds to the value 0
length(v[v <= 0])/length(v)
```

```
## [1] 0.847
```

```
# Theoretical percentile of 0 for the normal distribution (mean = -10, sd = 10)
pnorm(0, mean = -10, sd = 10)
```

```
## [1] 0.84134474607
```

- Create a function `my_reverse` which takes as required input a vector `v` and returns the vector in reverse where the first entry is the last entry, etc. No function calls are allowed inside your function otherwise that would defeat the purpose of the exercise! (Yes, there is a base R function that does this called `rev`). Use `head` on `v` and `tail` on `my_reverse(v)` to verify it works.

```
my_reverse = function(x){
  x = x[length(x):1]
}
```

```
head(my_reverse(v))
```

```
## [1] -2.29309174929 -24.20596983916 -14.47569927239 -29.90721019431
## [5]  0.50509338981  -4.22280360294
```

```
tail(my_reverse(v))
```

```
## [1]  2.9235484544  1.5809845504 -12.1287816897 -11.0797133222  -5.3128495996
## [6] -0.1803058865
```

- Create a function `flip_matrix` which takes as required input a matrix, an argument `dim_to_rev` that returns the matrix with the rows in reverse order or the columns in reverse order depending on the `dim_to_rev` argument. Let the default be the dimension of the matrix that is greater.

```
flip_matrix = function(X, dim_to_rev = NULL){
  if(is.null(dim_to_rev)){
    dim_to_rev = ifelse(nrow(X)>ncol(X),1,2)
  }

  if(dim_to_rev == 1){
    X = X[nrow(X):1, ]
  } else if(dim_to_rev == 2){
    X = X[, ncol(X):1]
  } else {
    stop("Invalid input: X is a square matrix!")
  }
}
```

```
D = matrix(data = seq(1:24), nrow = 4, ncol = 6,)
```

```
E = flip_matrix(D)
```

- Create a list named `my_list` with keys “A”, “B”, ... where the entries are arrays of size 1, 2 x 2, 3 x 3 x 3, etc. Fill the array with the numbers 1, 2, 3, etc. Make 8 entries according to this sequence.

```
my_list = list()
```

```
my_list$"A" = array(1, dim = 1)
my_list$"B" = array(1:2^2, dim = rep(2,2))
my_list$"C" = array(1:3^3, dim = rep(3,3))
my_list$"D" = array(1:4^4, dim = rep(4,4))
my_list$"E" = array(1:5^5, dim = rep(5,5))
my_list$"F" = array(1:6^6, dim = rep(6,6))
my_list$"G" = array(1:7^7, dim = rep(7,7))
my_list$"H" = array(1:8^8, dim = rep(8,8))
```

```
length(my_list)
```

```
## [1] 8
```

```
names(my_list)
```

```
## [1] "A" "B" "C" "D" "E" "F" "G" "H"
```

Run the following code:

```
lapply(my_list, object.size)
```

```
## $A
## 224 bytes
##
## $B
## 232 bytes
##
## $C
## 352 bytes
##
## $D
## 1248 bytes
##
## $E
## 12744 bytes
##
## $F
## 186864 bytes
##
## $G
## 3294416 bytes
##
## $H
## 67109104 bytes
```

```
?object.size
```

Use `?object.size` to read about what these functions do. Then explain the output you see above. For the later arrays, does it make sense given the dimensions of the arrays?

#TO-DO

Packages

Install the package `pacman` using regular base R.

```
#install.packages("pacman")
```

First, install the package `testthat` (a widely accepted testing suite for R) from <https://github.com/r-lib/testthat> using `pacman`. If you are using Windows, this will be a long install, but you have to go through it for some of the stuff we are doing in class. LINUX (or MAC) is preferred for coding. If you can't get it to work, install this package from CRAN (still using `pacman`), but this is not recommended long term.

```
pacman::p_load(testthat)
```

- Create vector `v` consisting of all numbers from -100 to 100 and test using the second line of code su

```
v = seq(-100, 100)
expect_equal(v, -100 : 100)
```

If there are any errors, the `expect_equal` function will tell you about them. If there are no errors, then it will be silent.

Test the `my_reverse` function using the following code:

```
expect_equal(my_reverse(v), rev(v))
expect_equal(my_reverse(c("A", "B", "C")), c("C", "B", "A"))
```

A little about strings

- Use the `strsplit` function and `sample` to put the sentences in the string `lorem` below in random order. You will also need to manipulate the output of `strsplit` which is a list. You may need to learn basic concepts of regular expressions.

```
lorem = "Lorem ipsum dolor sit amet, consectetur adipiscing elit. Morbi posuere varius volutpat. Morbi  
sample(unlist(strsplit(lorem, split = "\\.")))
```

```
## [1] "Mauris at sodales augue"
## [2] "Morbi posuere varius volutpat"
## [3] "Morbi faucibus ligula id massa ultricies viverra"
## [4] "Donec vehicula sagittis nisi non semper"
## [5] "Curabitur est augue, congue eget quam in, scelerisque semper magna"
## [6] "Cras suscipit id nibh lacinia elementum"
## [7] "Integer dapibus mi lectus, eu posuere arcu ultricies in"
## [8] "Lorem ipsum dolor sit amet, consectetur adipiscing elit"
## [9] "Aenean nulla ante, iaculis sed vehicula ac, finibus vel arcu"
## [10] "Donec at tempor erat"
```

You have a set of names divided by gender (M / F) and generation (Boomer / GenX / Millennial):

- M / Boomer “Theodore, Bernard, Gene, Herbert, Ray, Tom, Lee, Alfred, Leroy, Eddie”
- M / GenX “Marc, Jamie, Greg, Darryl, Tim, Dean, Jon, Chris, Troy, Jeff”
- M / Millennial “Zachary, Dylan, Christian, Wesley, Seth, Austin, Gabriel, Evan, Casey, Luis”
- F / Boomer “Gloria, Joan, Dorothy, Shirley, Betty, Dianne, Kay, Marjorie, Lorraine, Mildred”
- F / GenX “Tracy, Dawn, Tina, Tammy, Melinda, Tamara, Tracey, Colleen, Sherri, Heidi”
- F / Millennial “Samantha, Alexis, Brittany, Lauren, Taylor, Bethany, Latoya, Candice, Brittney, Cheyenne”

Create a list-within-a-list that will intelligently store this data.

```
#HINT:
#strsplit("Theodore, Bernard, Gene, Herbert, Ray, Tom, Lee, Alfred, Leroy, Eddie", split = ", ")[[1]]
#TO-DO

list_within_list = list()

list_within_list$M = list(
  Boomer = strsplit("Theodore, Bernard, Gene, Herbert, Ray, Tom, Lee, Alfred, Leroy, Eddie", split = ", ")
  GenX = strsplit("Theodore, Bernard, Gene, Herbert, Ray, Tom, Lee, Alfred, Leroy, Eddie", split = ", ")
  Millennial = strsplit("Theodore, Bernard, Gene, Herbert, Ray, Tom, Lee, Alfred, Leroy, Eddie", split = ", ")
)

list_within_list$F = list(
  Boomer = strsplit("Theodore, Bernard, Gene, Herbert, Ray, Tom, Lee, Alfred, Leroy, Eddie", split = ", ")
  GenX = strsplit("Theodore, Bernard, Gene, Herbert, Ray, Tom, Lee, Alfred, Leroy, Eddie", split = ", ")
  Millennial = strsplit("Theodore, Bernard, Gene, Herbert, Ray, Tom, Lee, Alfred, Leroy, Eddie", split = ", ")
)

list_within_list
```

```
## $M
## $M$Boomer
## [1] "Theodore" "Bernard" "Gene" "Herbert" "Ray" "Tom"
## [7] "Lee" "Alfred" "Leroy" "Eddie"
##
## $M$GenX
## [1] "Theodore" "Bernard" "Gene" "Herbert" "Ray" "Tom"
## [7] "Lee" "Alfred" "Leroy" "Eddie"
##
## $M$Millenial
## [1] "Theodore" "Bernard" "Gene" "Herbert" "Ray" "Tom"
## [7] "Lee" "Alfred" "Leroy" "Eddie"
##
##
## $F
## $F$Boomer
## [1] "Theodore" "Bernard" "Gene" "Herbert" "Ray" "Tom"
## [7] "Lee" "Alfred" "Leroy" "Eddie"
##
## $F$GenX
## [1] "Theodore" "Bernard" "Gene" "Herbert" "Ray" "Tom"
## [7] "Lee" "Alfred" "Leroy" "Eddie"
##
## $F$Millenial
## [1] "Theodore" "Bernard" "Gene" "Herbert" "Ray" "Tom"
## [7] "Lee" "Alfred" "Leroy" "Eddie"
list_within_list$M$GenX[3]

## [1] "Gene"
```

Now cleanup the namespace by deleting all stored objects and functions:

```
rm()
```